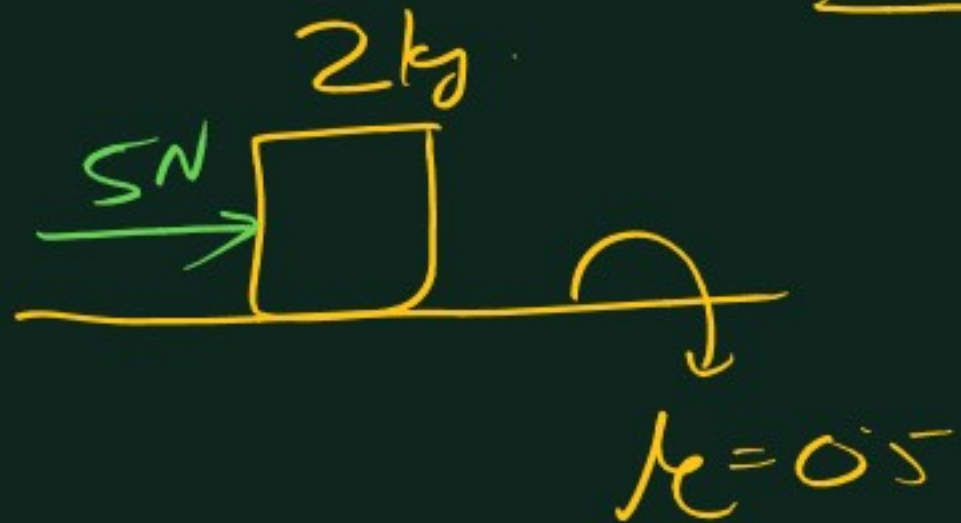


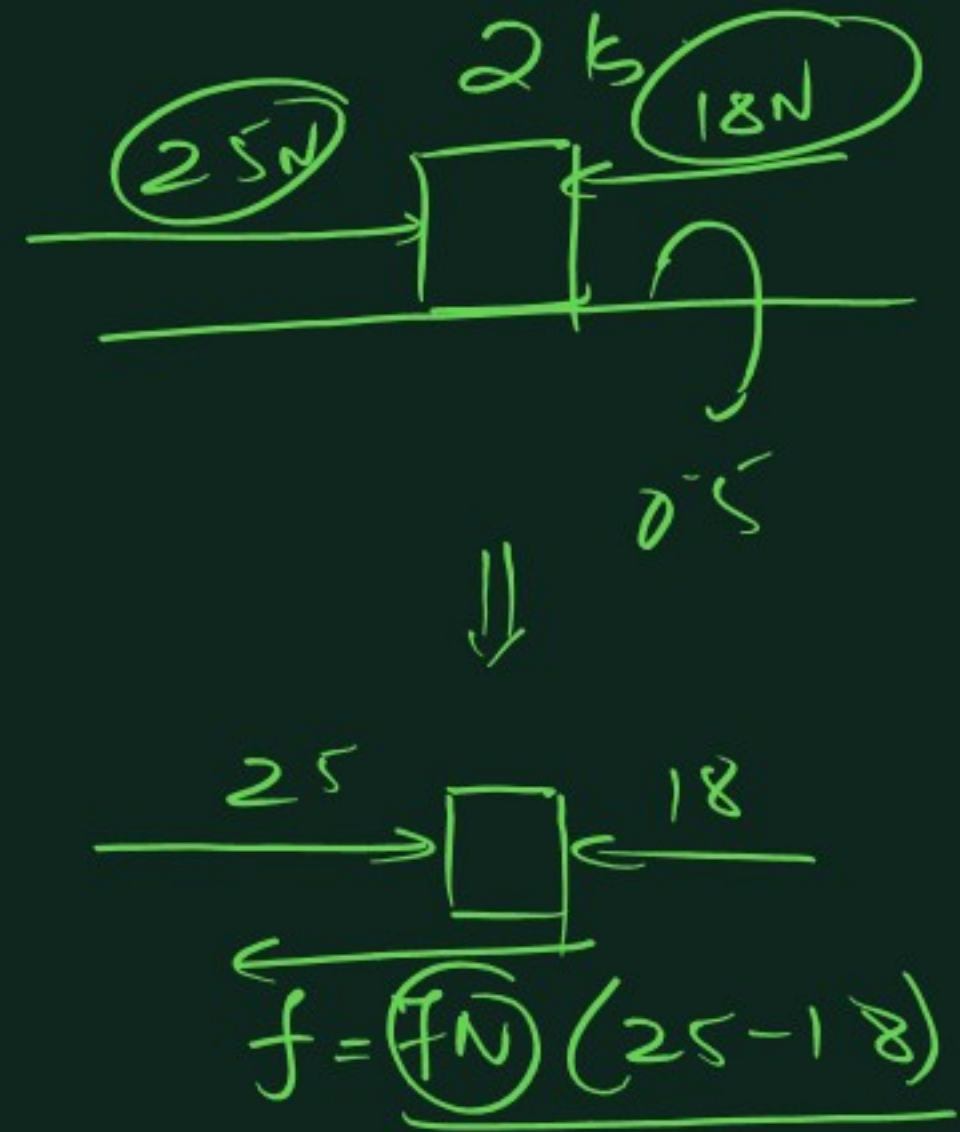
Kinetic $f \rightarrow \mu_k N$

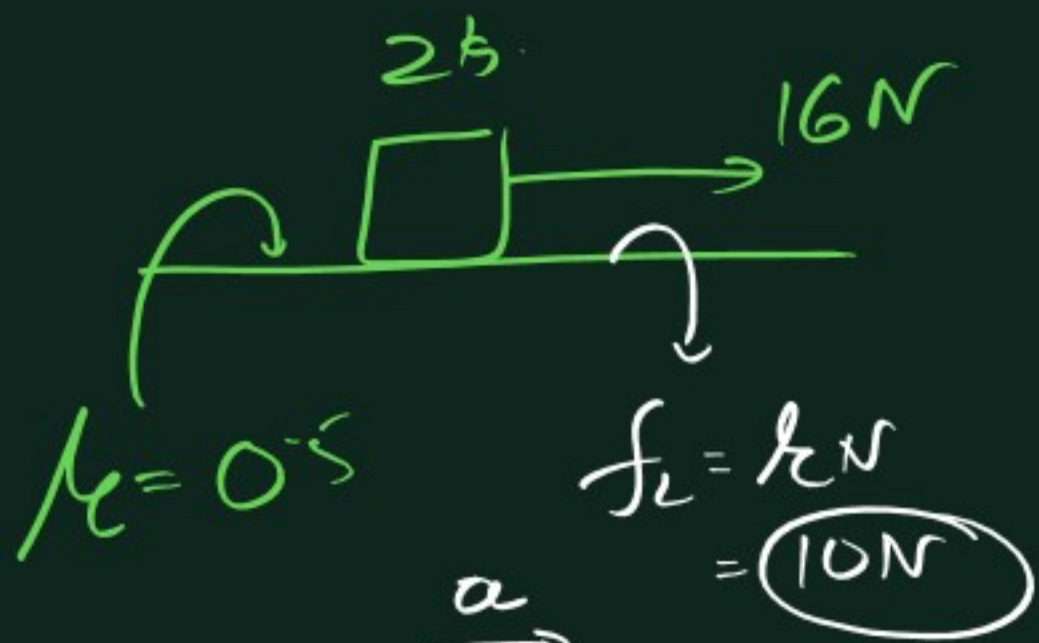


$$\begin{aligned} f_L &= \mu N \\ &= \mu mg \\ &= 0.5 \times 20 = 10 \end{aligned}$$

$$f = \underline{5N}$$

$$0 < f \leq f_L$$

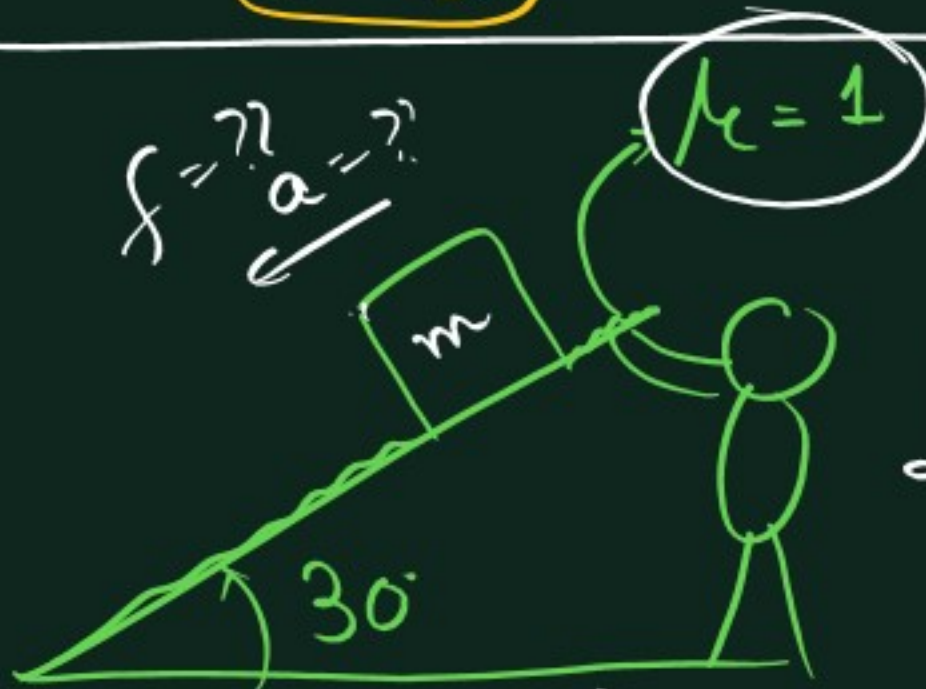




$f = 10 \text{ N}$
 $\Sigma F = 16 - 10 = 2 \times a$
 $\Rightarrow a = \frac{6}{2} = 3$



$\tan \theta = \frac{1}{\sqrt{3}}$
 $\Rightarrow \theta = 30^\circ$



$\tan 30^\circ = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3} \approx 0.6$

$a = 0$

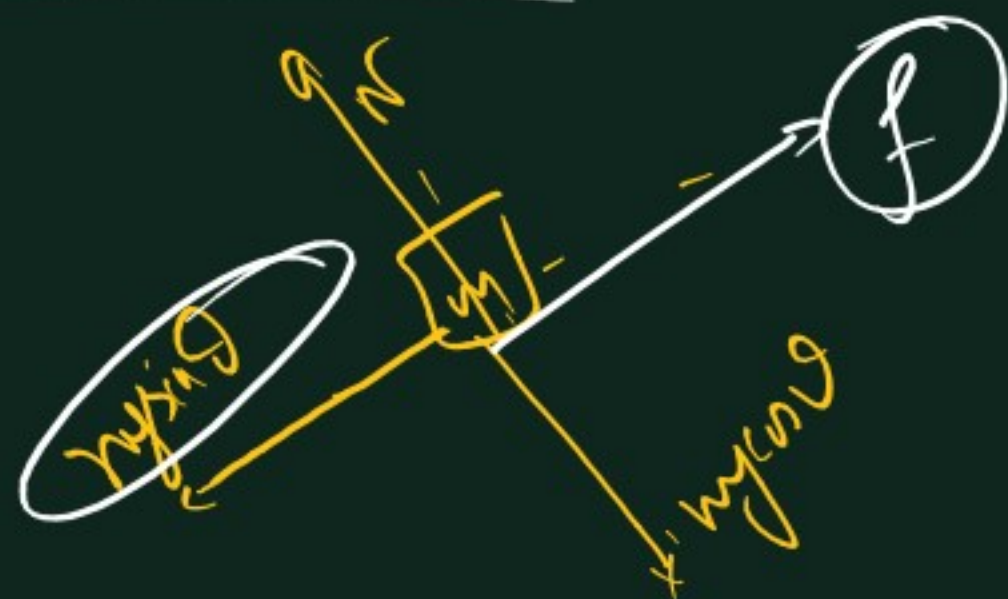
$\tan \theta < \mu$

$\tan \theta, \mu$

If $\tan \theta = \mu$
 $\Rightarrow$ Limiting case

If $\tan \theta > \mu$
 $\Rightarrow$ Slide

$\tan \theta < \mu$
 $\Rightarrow$ Rest

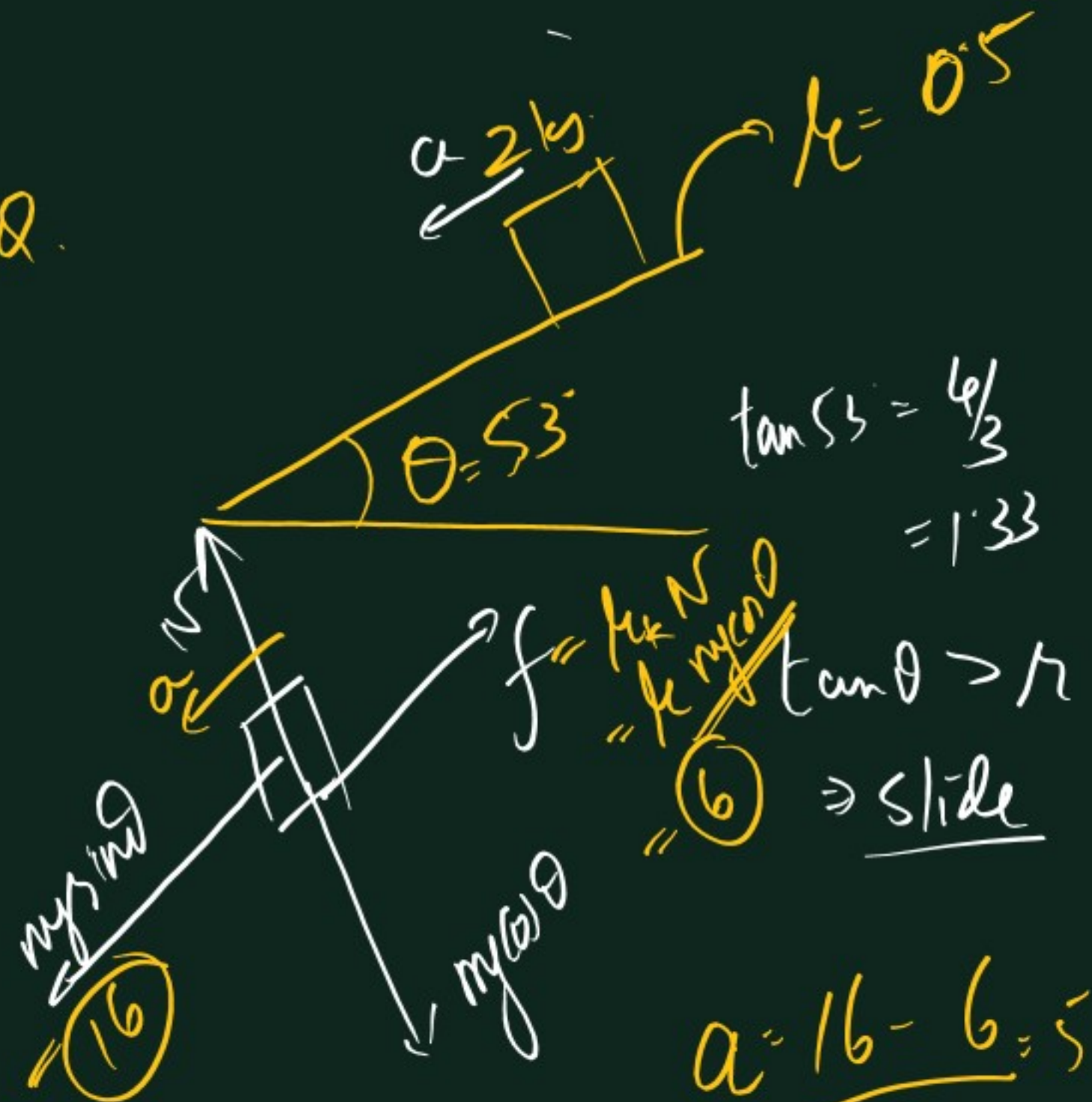


as it is at rest

$$\sum F_x = 0$$

$$\Rightarrow f = mg \sin \theta$$

Q.



$$\tan 53 = \frac{4}{3} = 1.33$$

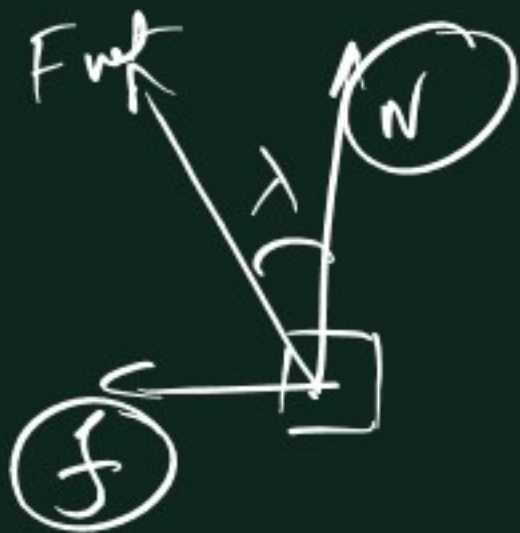
$$f = \mu N = \mu mg \cos \theta$$

$$6 = 0.5 \times 16 \times \cos 53$$

$$\Rightarrow \text{slide}$$

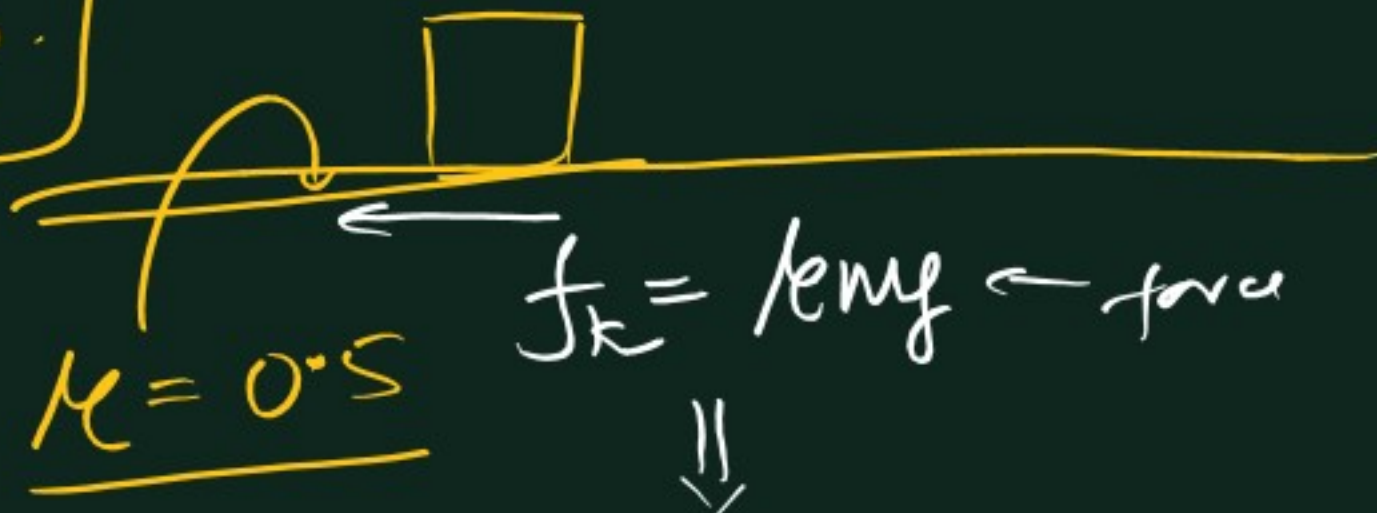
$$a = \frac{16 - 6}{2} = 5$$

Angle of $f \rightarrow \theta$ b/w N & F_{net} when just about to slip.



$$\tan \lambda = \mu$$

20 m/s ← speed



$$f_k = \mu mg \leftarrow \text{force}$$

$$a = \frac{\mu mg}{m} = \mu g$$

$\mu = 5$



Kinematics

$$u = 20$$

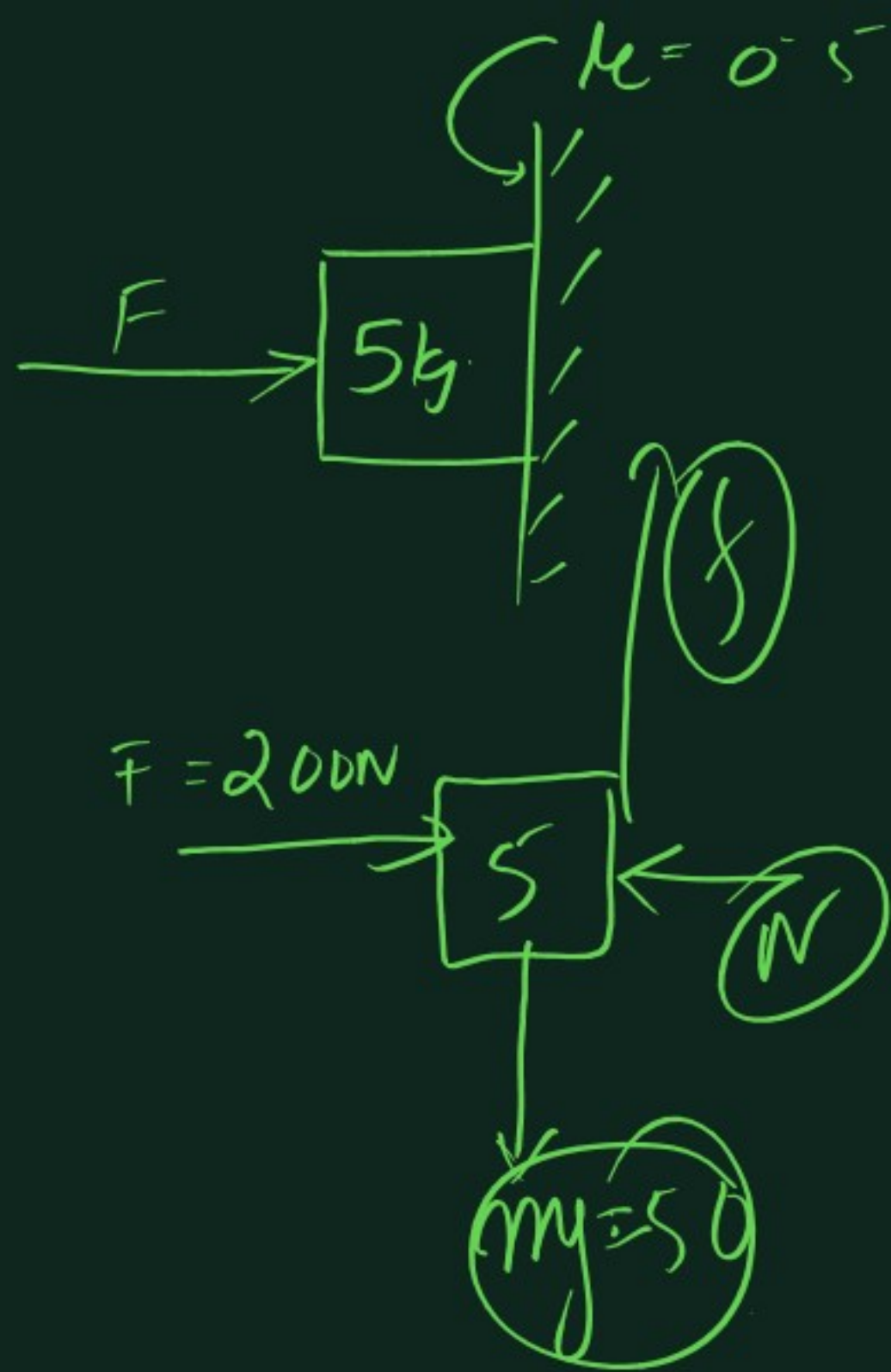
$$a = -5$$

$$v = 0$$

$$s = \left| \frac{u^2}{2a} \right|$$

$= 40$

$$v^2 = u^2 + 2as$$



$$N = 200$$

$$f_L = \mu N = 0.5 \times 200 = \underline{\underline{100}}$$

$$f = mg = 50\text{N}$$

Intext Q

3)



A) 20°

C) 45°

$\tan 45 = 1$

$\tan \theta = h$

B) $30^\circ \rightarrow \tan 30 = \frac{1}{\sqrt{3}}$
 $= \underline{0.577}$

D) 15°

3) 3 in 5



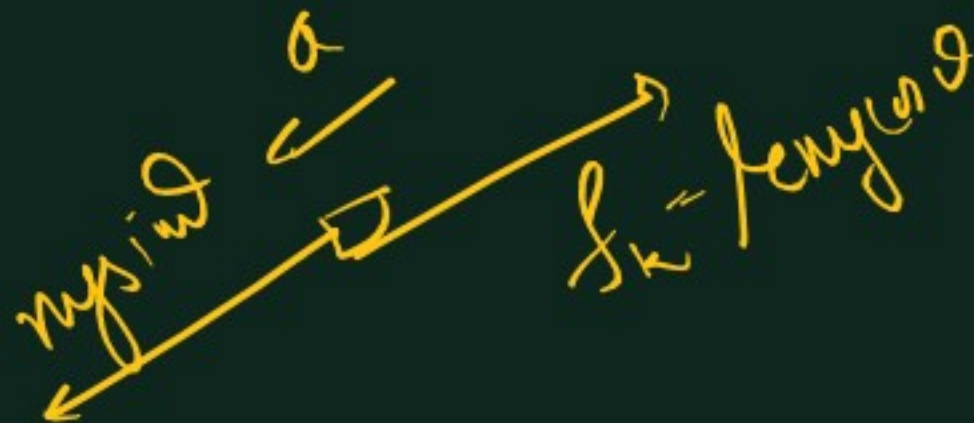
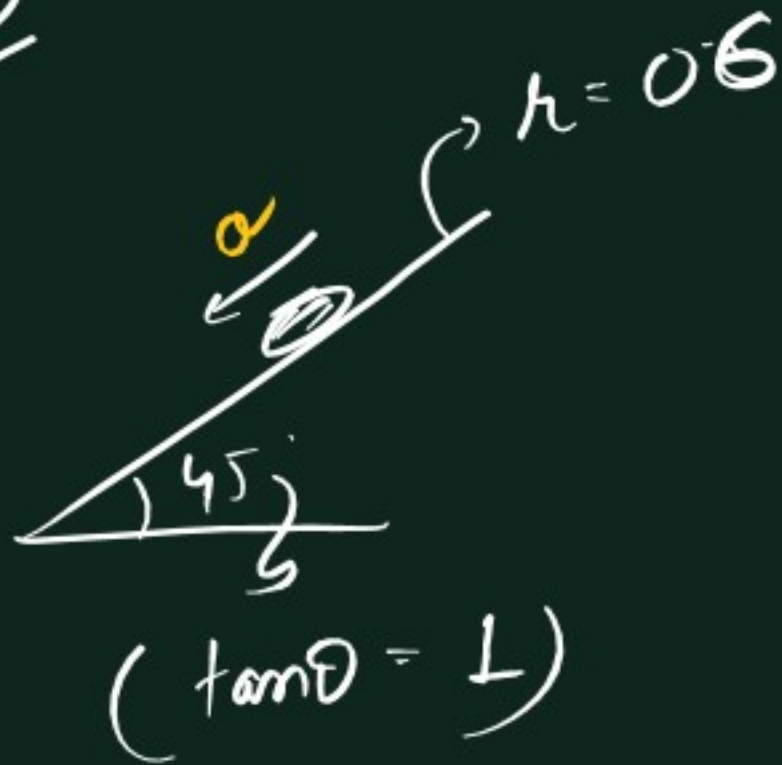
$\sin \theta = \frac{3}{5} (\theta = 37^\circ)$

$\Downarrow$

$\tan \theta = \frac{3}{4}$
 $= \underline{0.75}$

$\tan \theta = h$

Intext 2
17)



$$\Sigma F = ma$$

$$\Rightarrow mg \sin \theta - \mu mg \cos \theta = ma \Rightarrow a = g \sin \theta - \mu g \cos \theta$$

Intext 3

$$\mu = 0.75$$

Angle of friction (λ)

$$\tan \lambda = \mu \rightarrow \frac{3}{4}$$

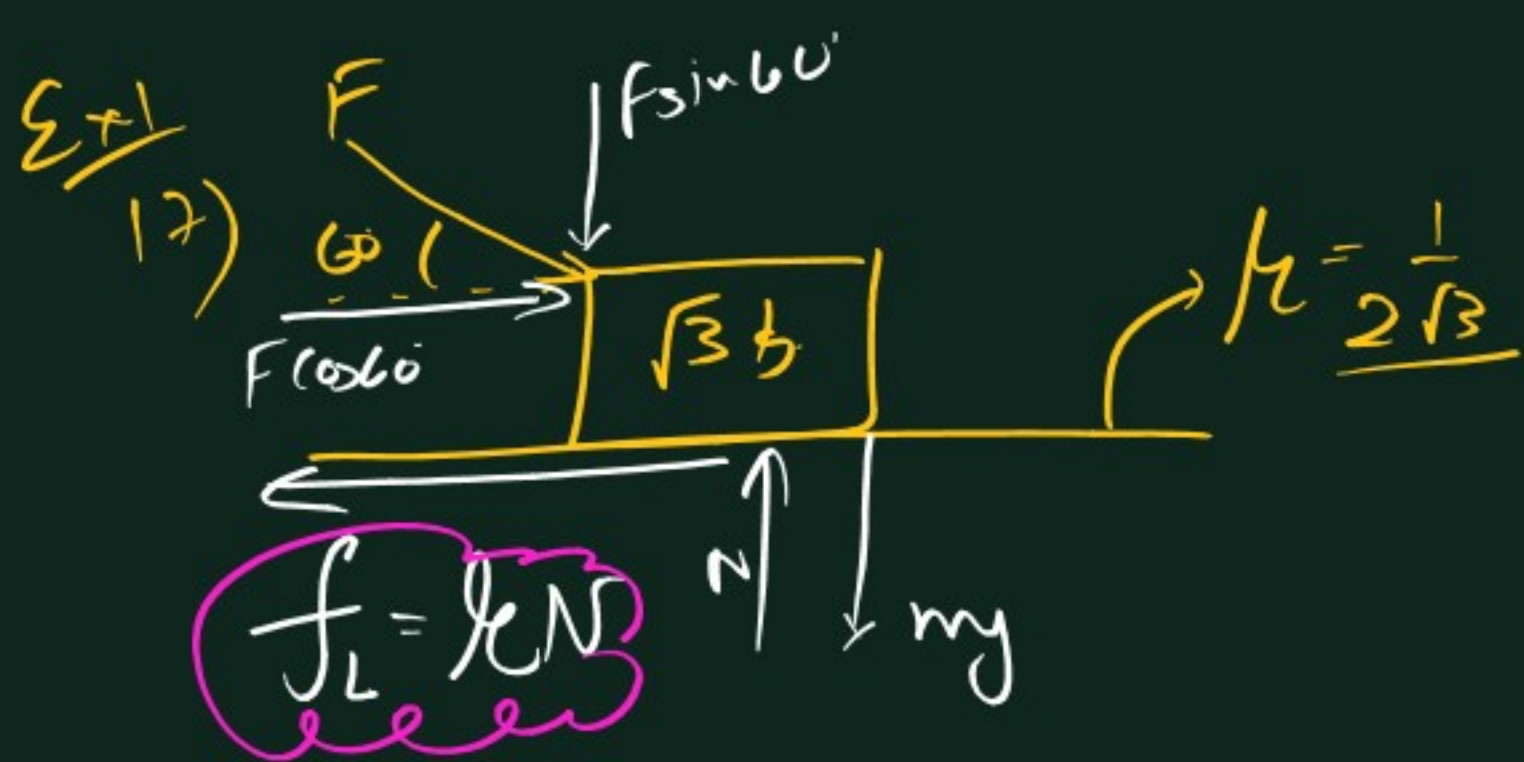
$$\Rightarrow \lambda = 37^\circ$$

$$\sin \theta = \frac{3}{5}$$

Find $\tan \theta$??

$$\frac{1}{2\sqrt{2}}$$





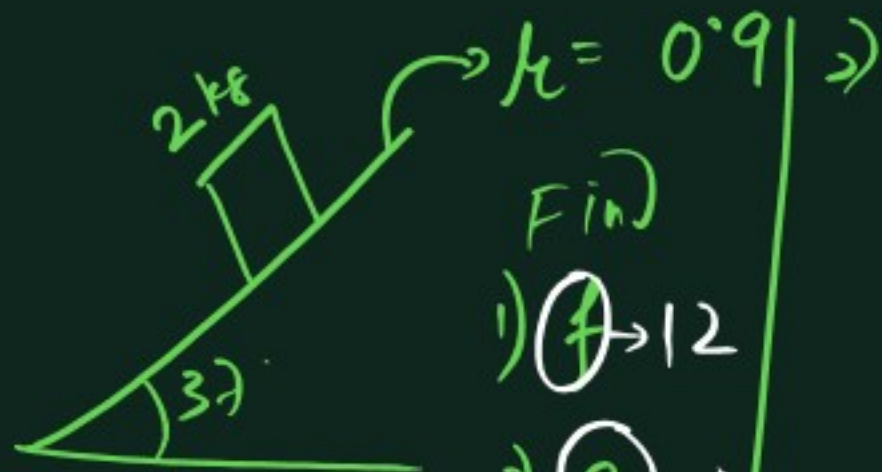
$$\sum F_y = 0$$

$$\Rightarrow N = mg + F \frac{\sqrt{3}}{2} \quad \text{--- (1)}$$

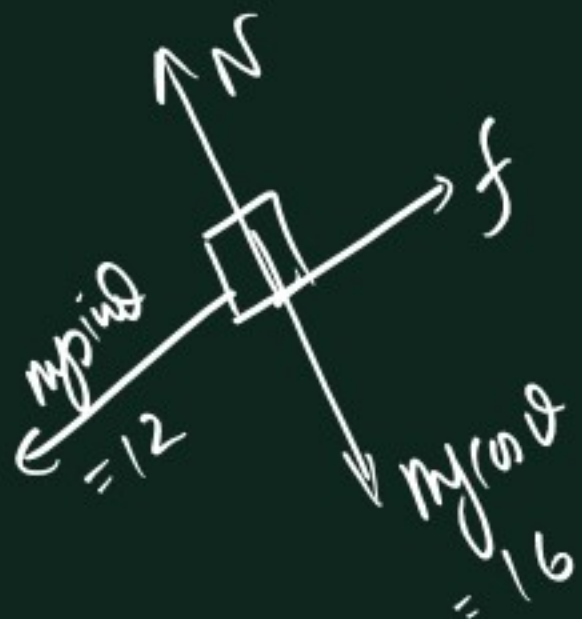
$$\sum F_x = 0$$

$$F \cos 60 = \mu N \Rightarrow (mg + F \frac{\sqrt{3}}{2})$$

Q1



Find
 1) $f \rightarrow 12$
 2) $a \rightarrow 0$

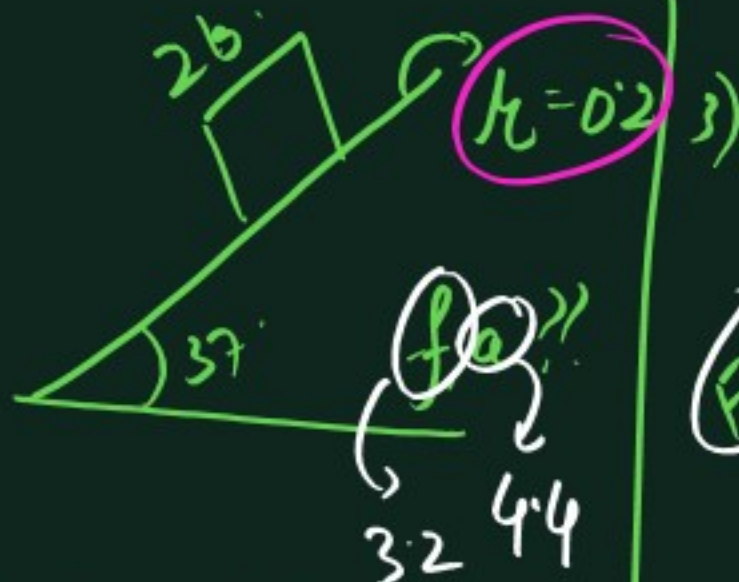


$$\rightarrow N = 16$$

$$\tan \theta = \frac{3}{4} = 0.75 < k \Rightarrow \text{Rest}$$

$$\therefore f = mg \sin \theta = 12$$

$$a = 0$$



$$\tan \theta = 0.75$$

$$k = 0.2$$

$$\tan \theta > k$$

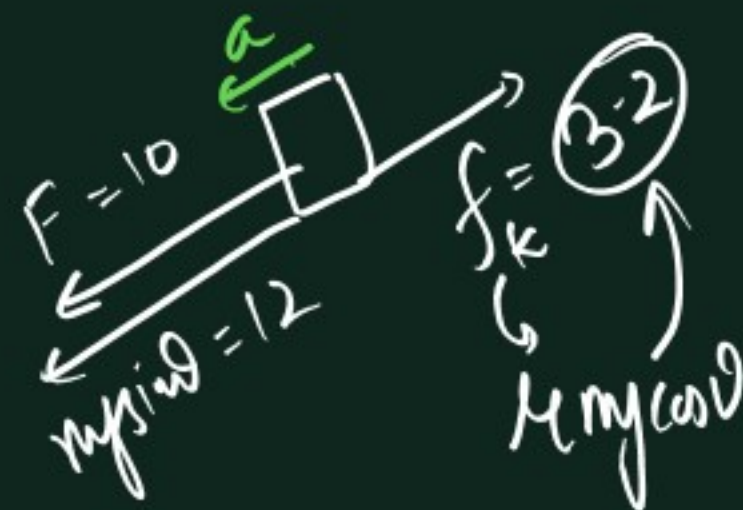
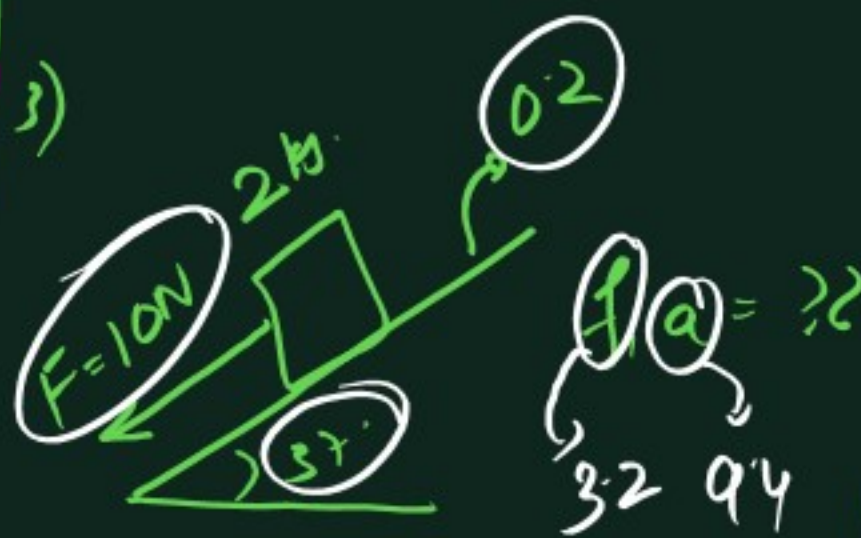
$$\text{Slide} \Rightarrow f_k = kN$$

$$= k mg \cos \theta$$

$$= 0.2 \times 16$$

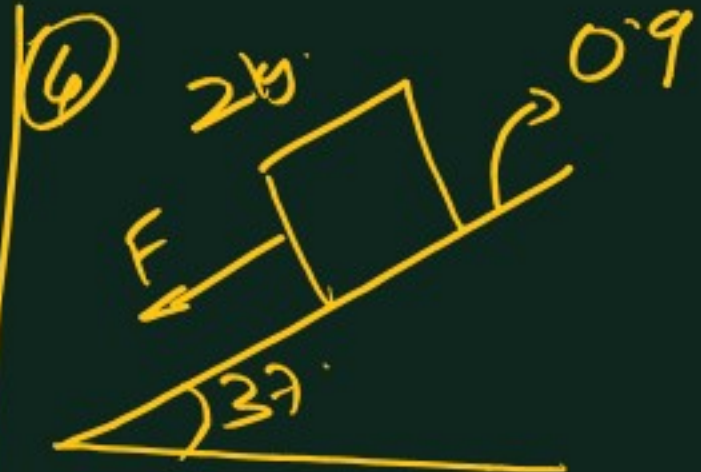
$$= 3.2$$

$$a = \frac{\sum F}{m} = \frac{12 - 3.2}{2} = 4.4$$



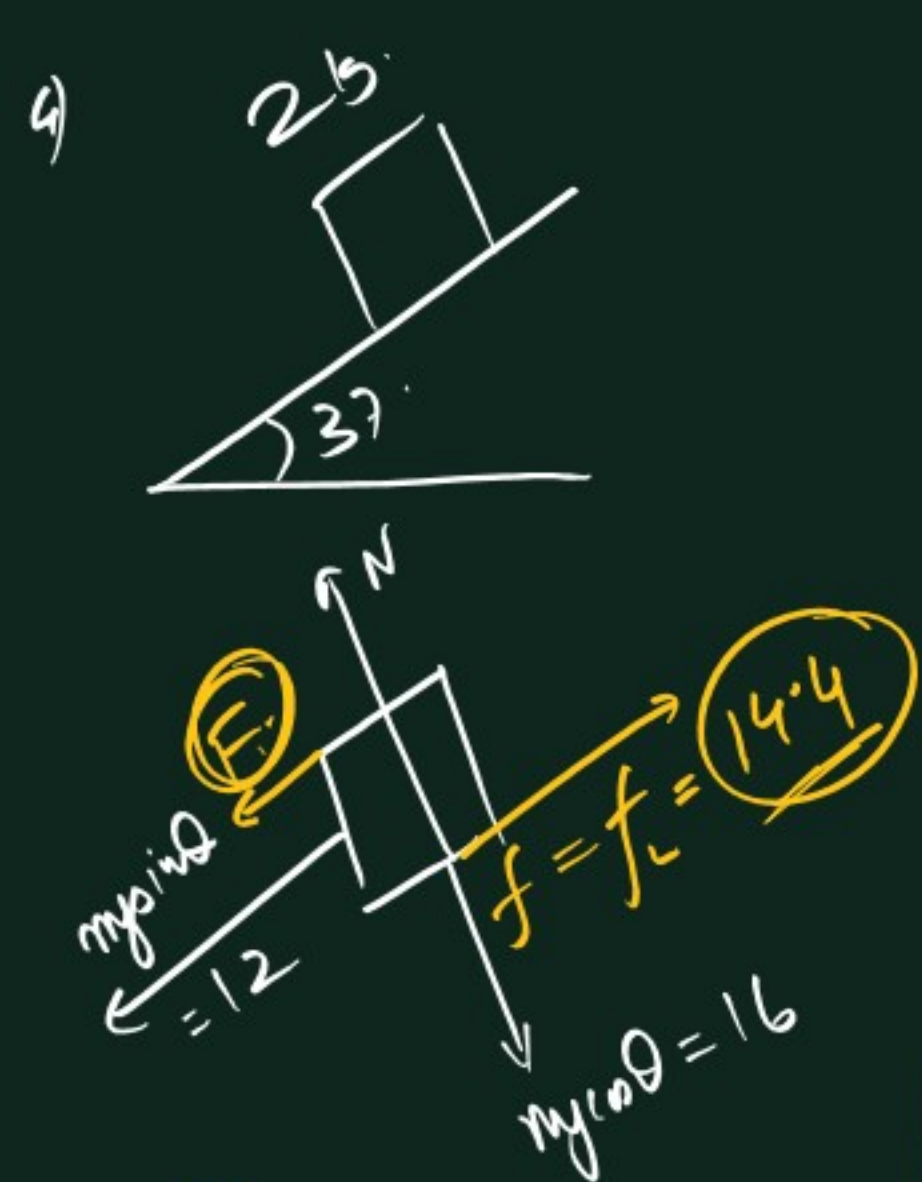
$$a = \frac{10 + 12 - 3.2}{2}$$

$$= 9.4$$



Min. F to slide the block?

- A) 14.4
- B) 2.4
- C) 8.8
- D) 3.4



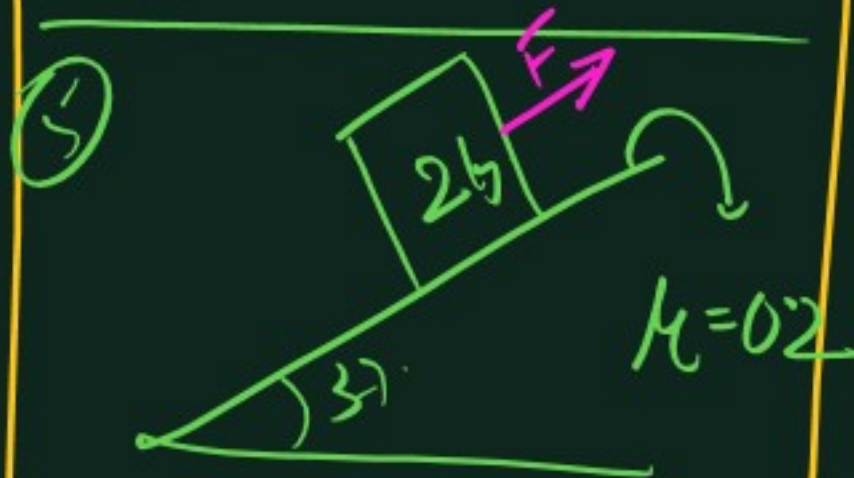
$$f_c = \mu_r mg \cos \theta$$

$$= 0.9 \times 16 = \underline{\underline{14.4}}$$

$$F_{\min} + mg \sin \theta = f_c$$

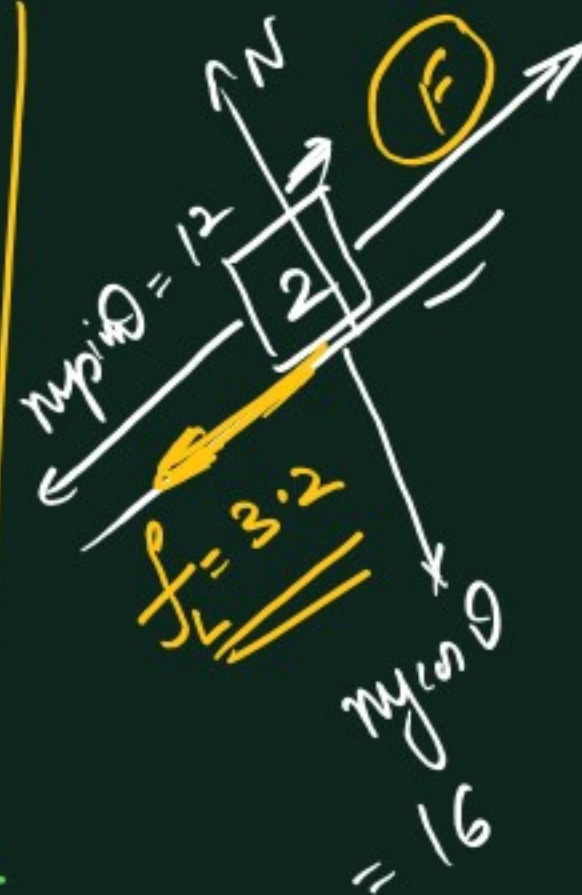
$$\Rightarrow F_{\min} = f_c - mg \sin \theta$$

$$= 14.4 - 12 = \underline{\underline{2.4}}$$



Min. F required to slide up the block.

$$\underline{\underline{15.2 \text{ N}}}$$



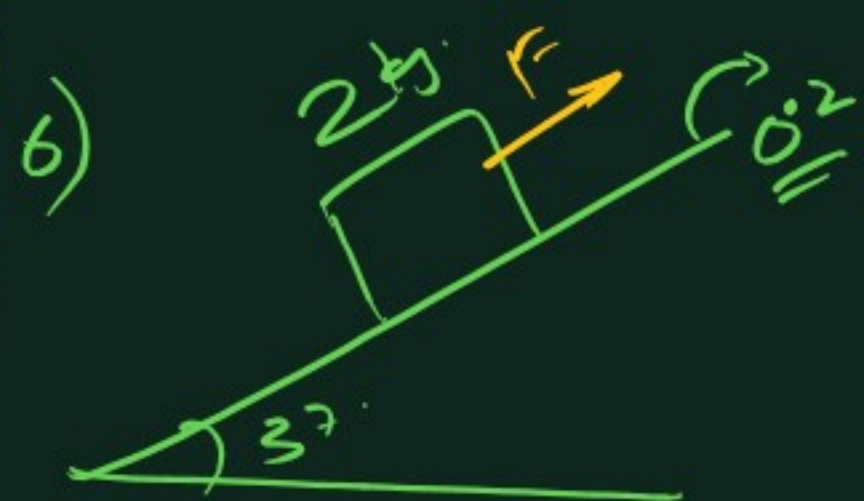
$$f_c = \mu N = 0.2 \times 16$$

$$= \underline{\underline{3.2}}$$

$$F = mg \sin \theta + (\mu mg \cos \theta)$$

$$= 12 + 3.2$$

$$= \underline{\underline{15.2 \text{ N}}}$$

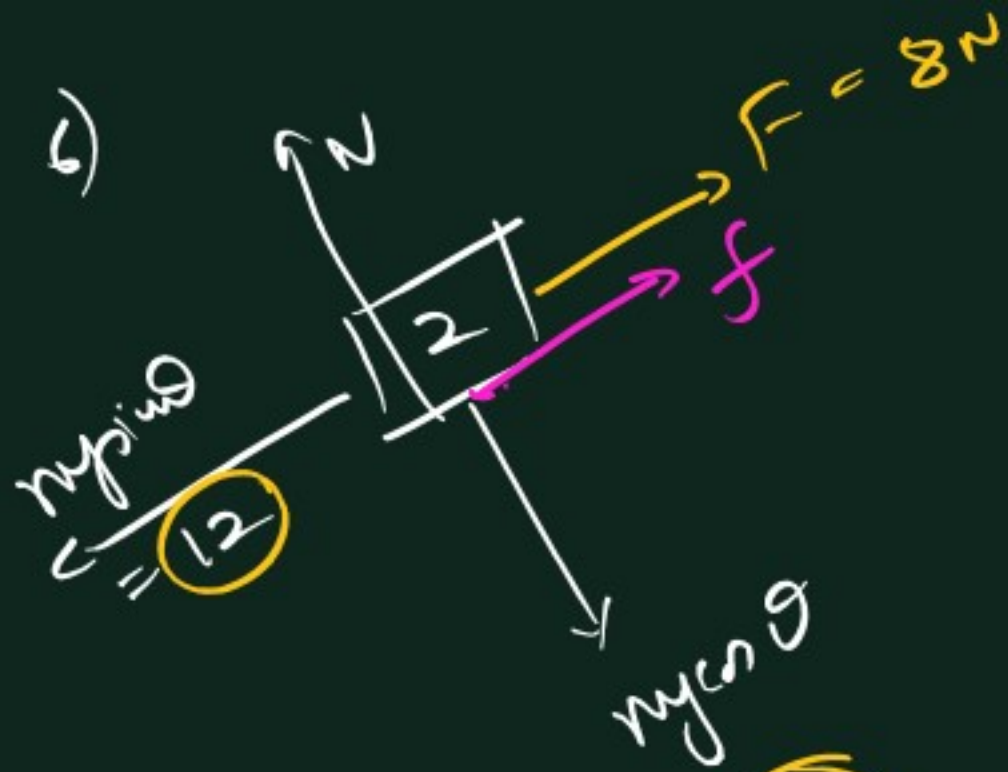


Find f if

i) $F = 8 \text{ N}$

ii) $F = 12 \text{ N}$

iii) $F = 15 \text{ N}$

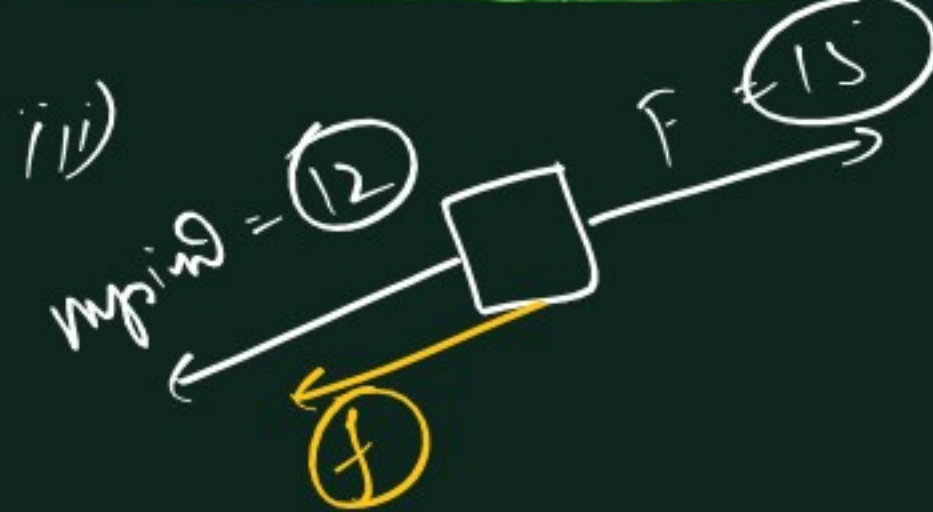
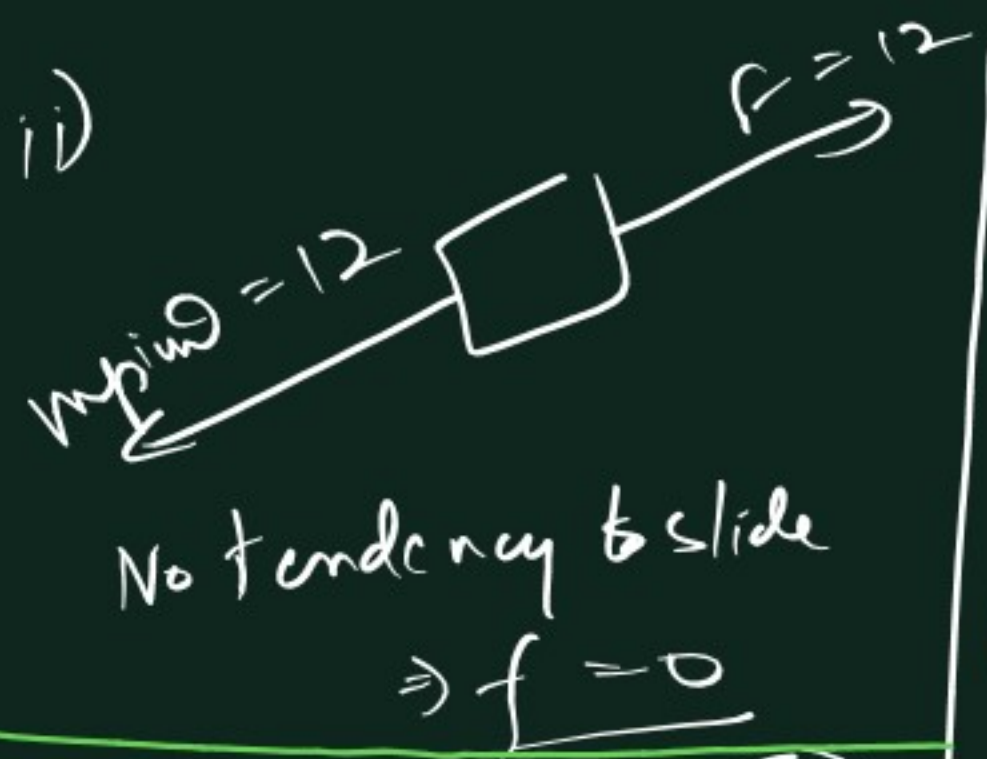


$$f_L = \mu mg \cos \theta = 3 \cdot 2 \text{ N}$$

To keep at rest
required $f = 12 - 8$
 $= 4 \text{ N}$

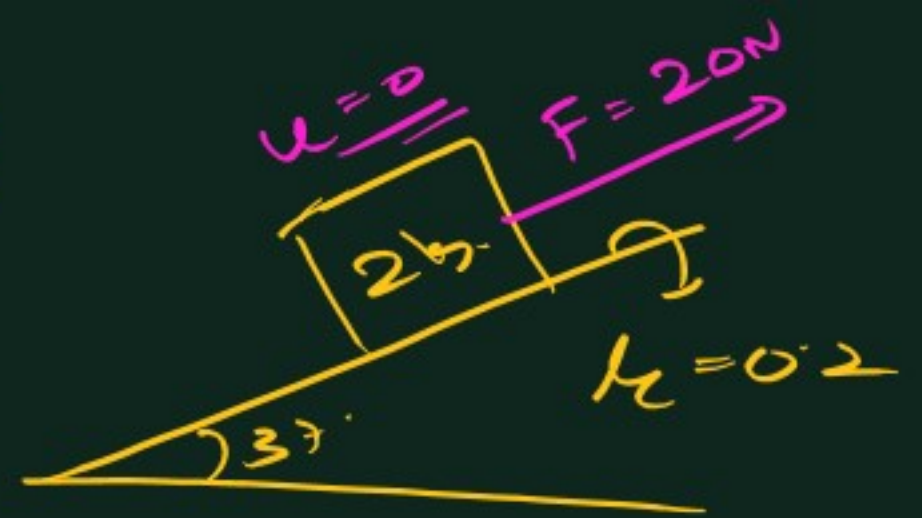
But $f_L = 3.2$

As $f_L < \text{Required } f$
 $\Rightarrow$ Sliding $\Rightarrow$ $f_{\text{kinetic}} = 3.2 \text{ N}$ acts



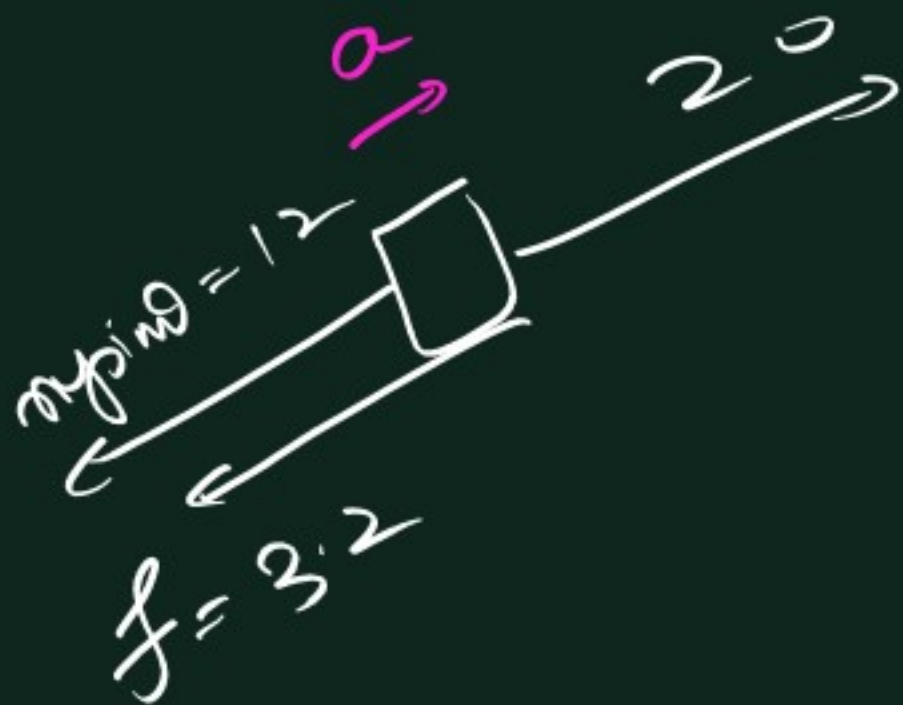
$$f = 15 - 12 = 3 \text{ N} < f_L$$

$\therefore f = 3 \text{ N}$ acts & block at rest



Find i) a

ii) time to cover 2m



$$\Sigma F = ma$$

$$\Rightarrow 20 - 12 - 3.2 = 2a$$

$$\Rightarrow \boxed{a = 2.4}$$

$$s = \cancel{ut} + \frac{1}{2}at^2$$

$$\Rightarrow 2 = \frac{1}{2} \times 2.4 \times t^2$$

$$\Rightarrow t = \sqrt{\frac{4}{2.4}}$$

$$= \sqrt{\frac{5}{3}}$$

$$= \frac{\sqrt{5}}{\sqrt{3}} = \sqrt{1.66}$$

$$= \sqrt{\frac{1.66}{100}} = \frac{\sqrt{166}}{10} = \frac{12.9}{10} = \underline{\underline{1.29s}}$$

$$\frac{2.24}{1.73} = \frac{2.24}{1.73}$$

=

$$\sqrt{\frac{7}{2}} = \sqrt{3.5}$$

=

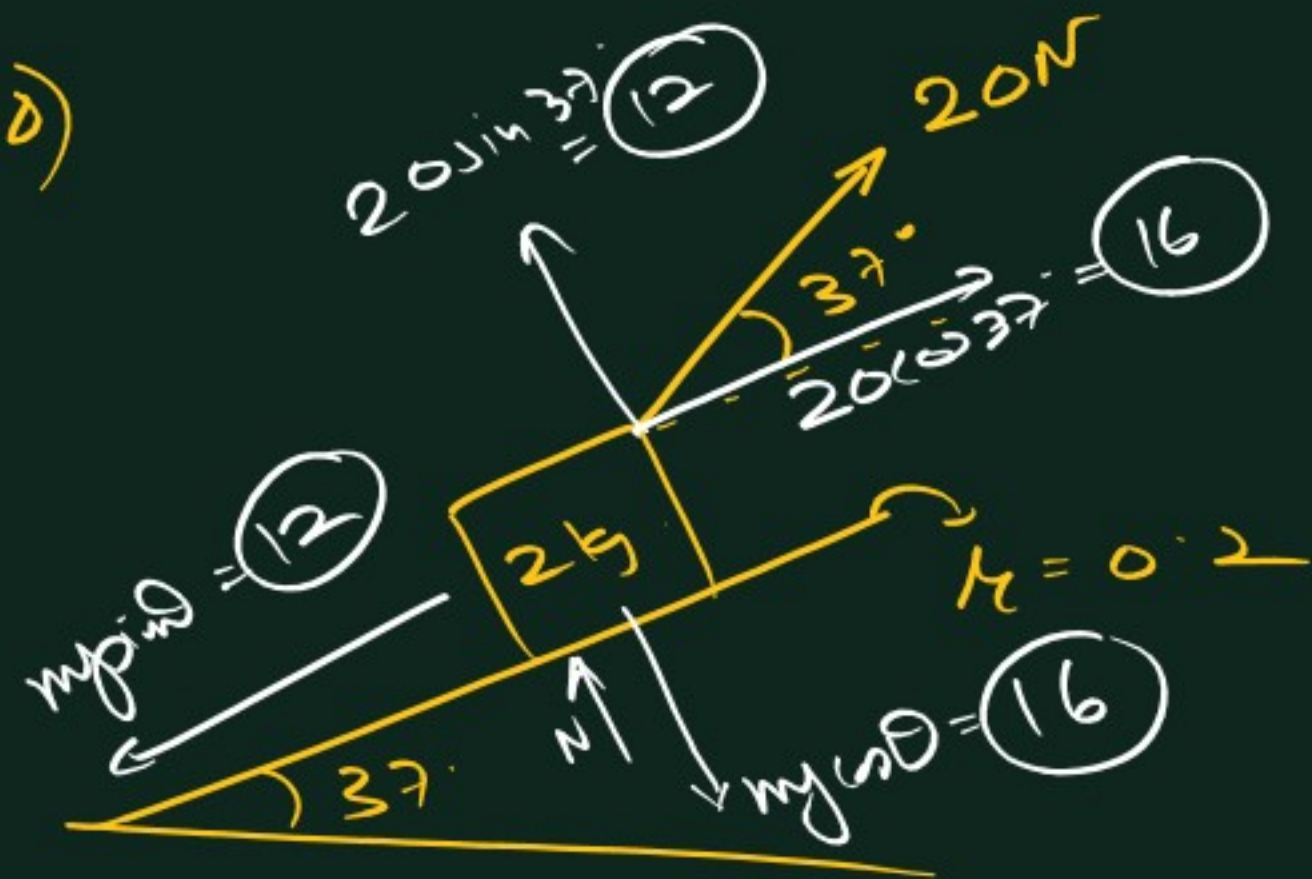
$$= \sqrt{\frac{350}{100}}$$

$$= \sqrt{3.5}$$

$$= \frac{18.7}{10}$$

$$= \underline{\underline{1.87}}$$

Q2)



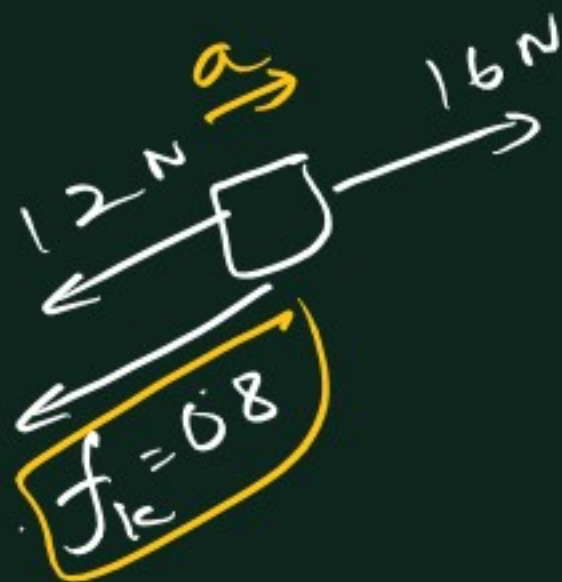
Find friction & a

$$\sum F_y = 0$$

$$N + 12 = 16$$

$$\Rightarrow \underline{N = 4}$$

$$\rightarrow f_k = \mu N = 0.2 \times 4 = 0.8$$



$$\sum F = ma$$

$$\Rightarrow 16 - 12 - 0.8 = 2a$$

$$\Rightarrow \boxed{a = 1.6}$$

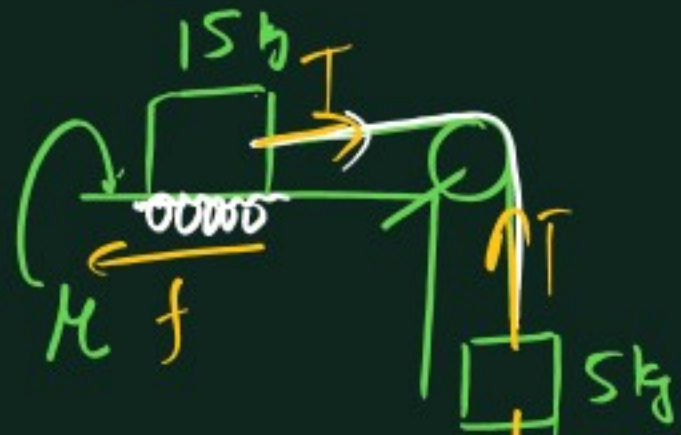
To stay at rest

$$f = 4N$$

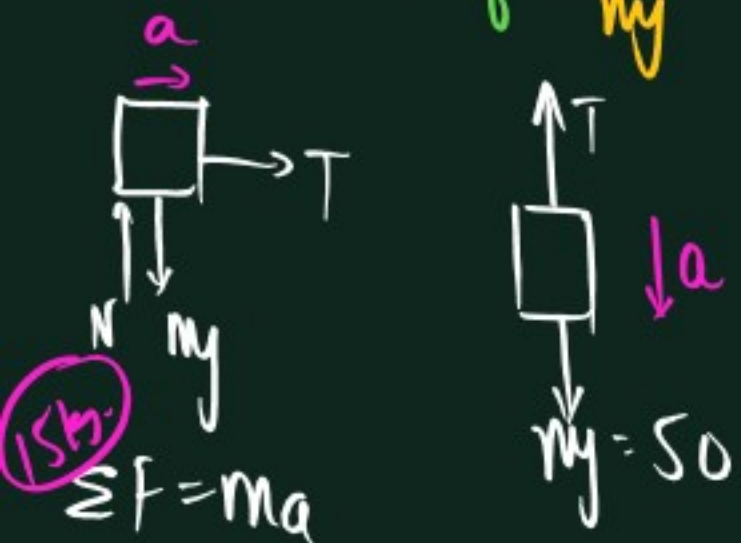
Not possible
($f_k = 0.8$)

$$\begin{cases} \bullet f = 4 \\ \bullet f = 0.8 \\ \bullet a = 1.6 \end{cases}$$

Pulley Block



Find a if $\mu = 0$



$\Sigma F = ma$
 $\Rightarrow T = 15a \text{ --- (1)}$

$50 - T = 5a \text{ --- (2)}$

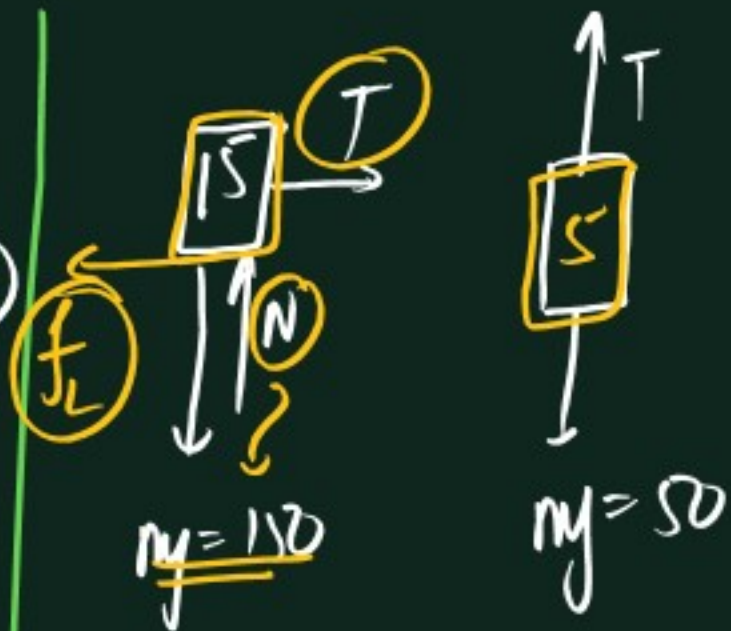
Add $a = 2.5 \text{ m/s}^2$

Q2

$a = \frac{F_{\text{net pulling}}}{\text{Total mass}}$
 $= \frac{50}{15+5} = 2.5$

2) In above question, find μ_{minimum} to keep system at rest.

$f = f_{\text{limiting}} = \mu N$

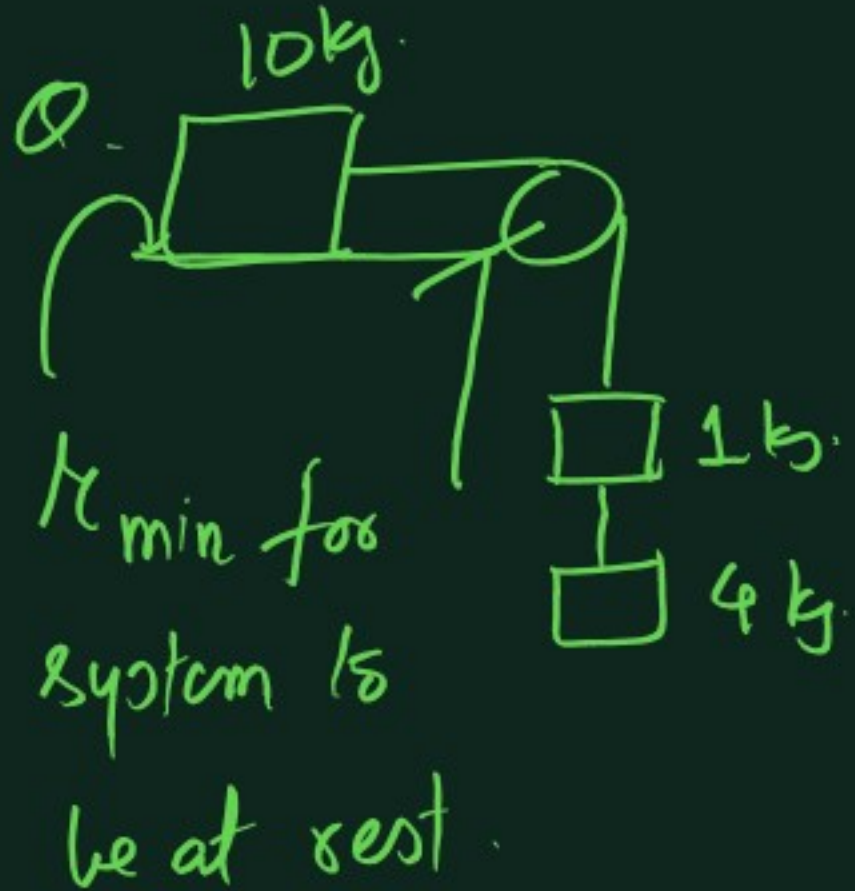


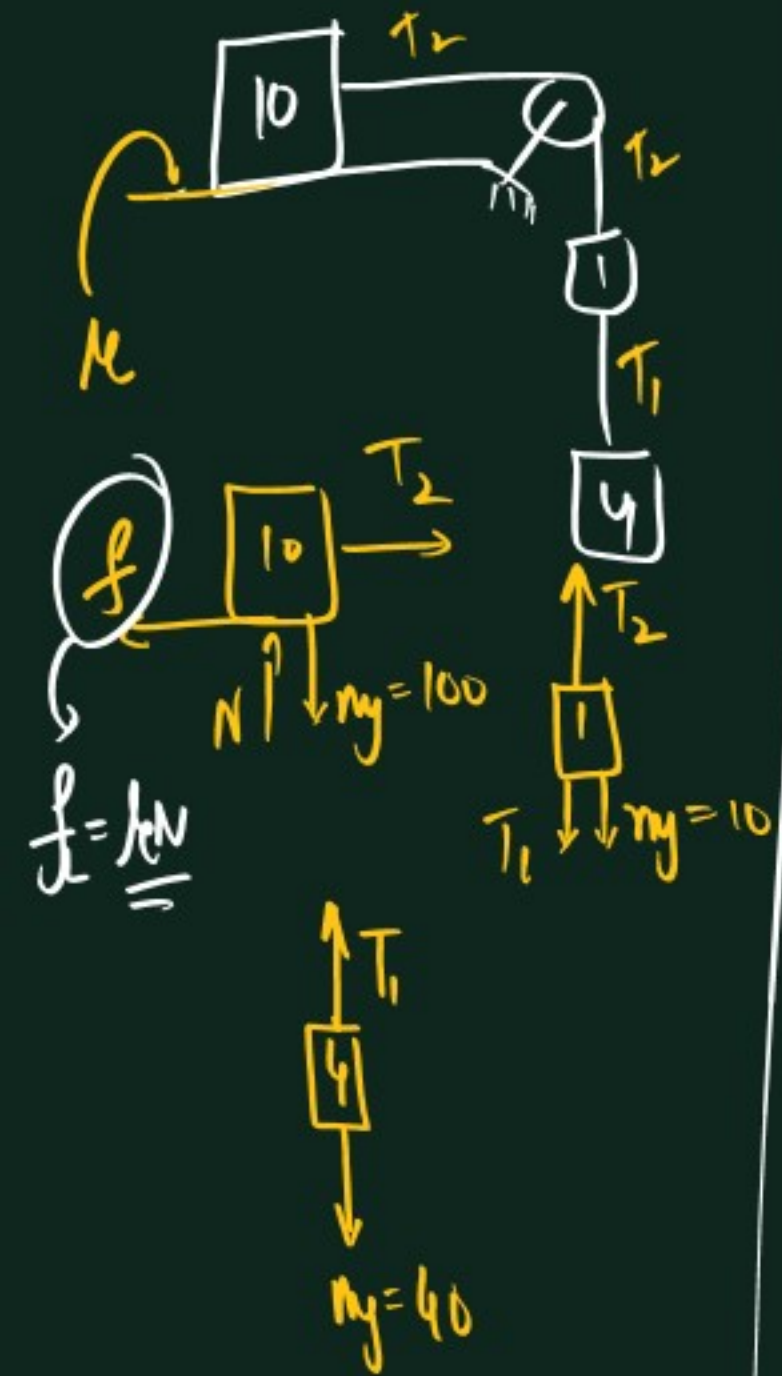
$\Sigma F = 0$

$50 - T = 0 \text{ --- (1)}$

$150 - T = f_L$
 $\Rightarrow 50 = \mu N$
 $= \mu \times 150$

$\Rightarrow \mu = \frac{1}{3}$





(4b)

$$T_1 = 40$$

(1b)

$$T_2 = T_1 + mg = 40 + 10 = 50$$

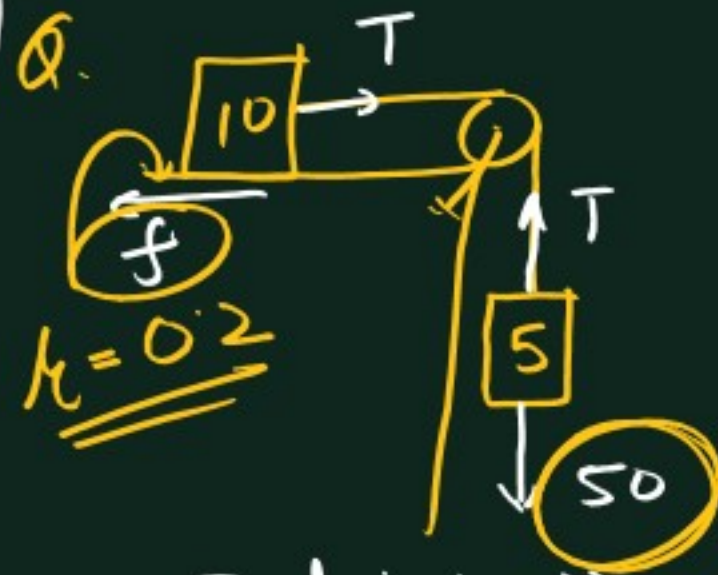
(10b)

$$T_2 = f_L$$

$$\Rightarrow 50 = \mu N \Rightarrow 100$$

$$\Rightarrow \mu = \frac{50}{100} = \frac{1}{2} = 0.5$$

Q.



Find $a = ??$

Step 1 Assume at rest

And find f

2 Compare f with f_L

3 If $f \leq f_L \Rightarrow$ Rest

$f > f_L \Rightarrow a$

Assuming at rest



$$T = 50 \quad \text{--- (1)}$$

$$T = f$$

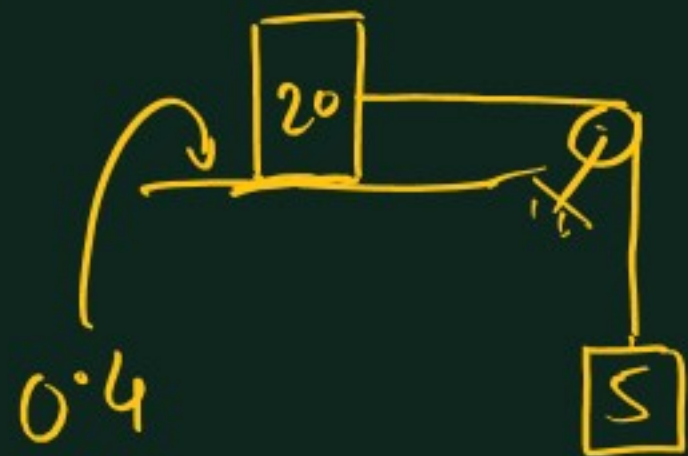
$$\Rightarrow f = 50$$

$$f_L = \mu N = \mu \times 100 = 20\text{ N}$$

System will a

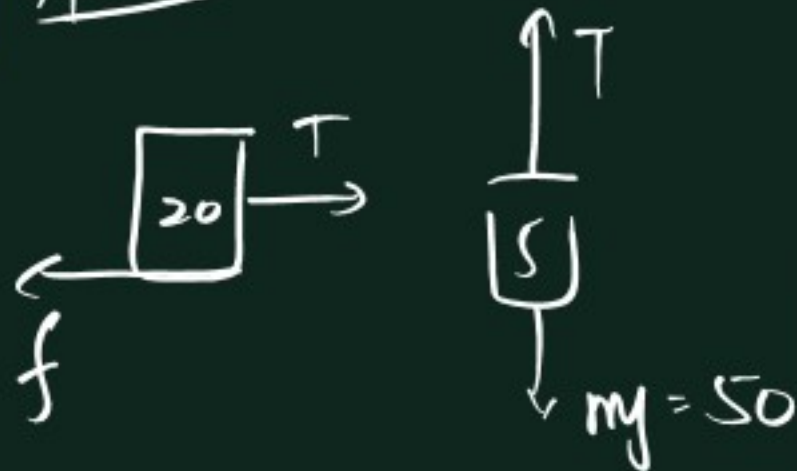
$$\therefore a = \frac{\text{Net Pulling F}}{\text{Total Mass}} = \frac{50 - 20}{15} = 2$$

Q.

Find a & f ??

- Q. A) $\frac{6}{5}$ B) 0
C) $\frac{26}{5}$ D) None

Assume rest



$$T = 50 \quad \text{--- (1)}$$

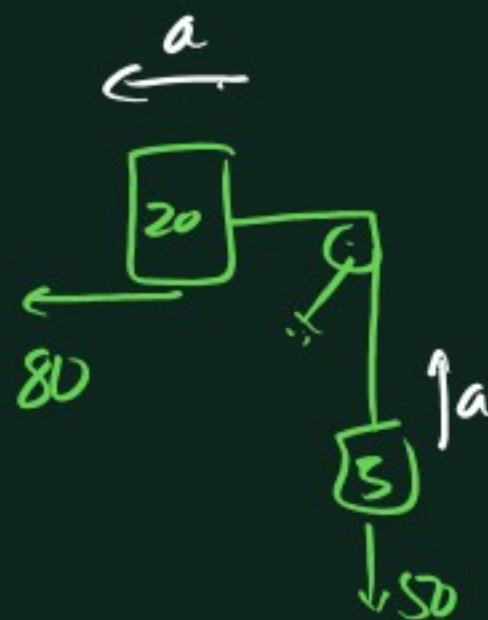
$$f = T$$

$$\Rightarrow \boxed{f = 50}$$

$$f_c = \mu_r (m_1 g) = \boxed{80 \text{ N}}$$

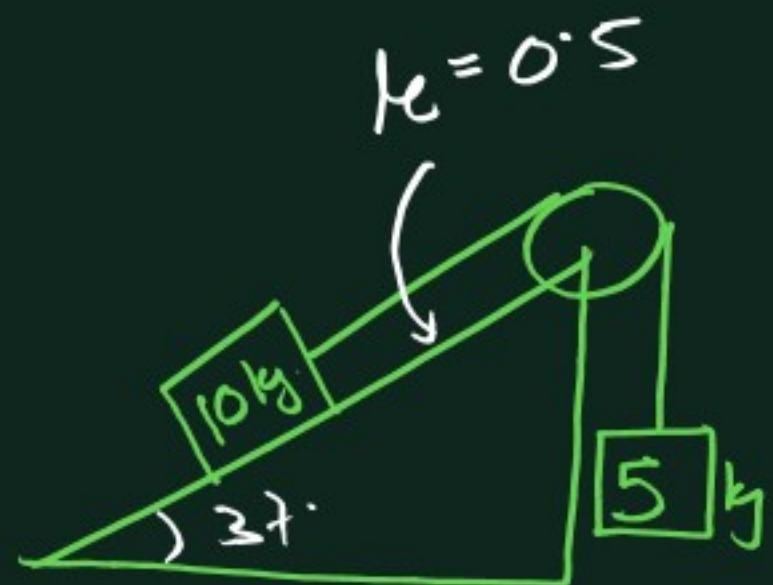
$\therefore$ System at rest $a = 0$
 $f = 50 \text{ N}$.

Mistake



$$a = \frac{80 - 50}{25} = \boxed{\left(\frac{6}{5}\right)}$$

Q.

 $a = ??$

- A) 2 (✓)
B)

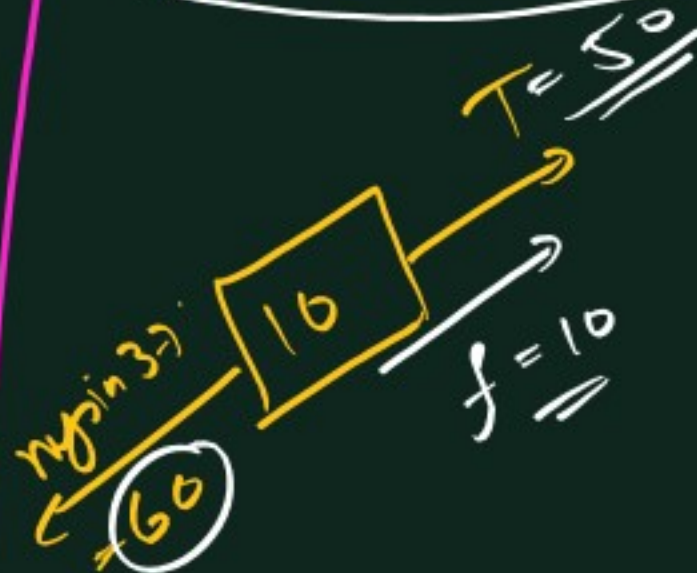
Ans:



A) 2 (→) B) 2 (←)

C) $\frac{4}{13}$ (→) D) 0

Assume at rest



5) $T = 50$

10) $f = 60 - 50 = 10$

$f_L = \mu N = \mu mg \cos 37^\circ = 0.5 \times 80 = 40$

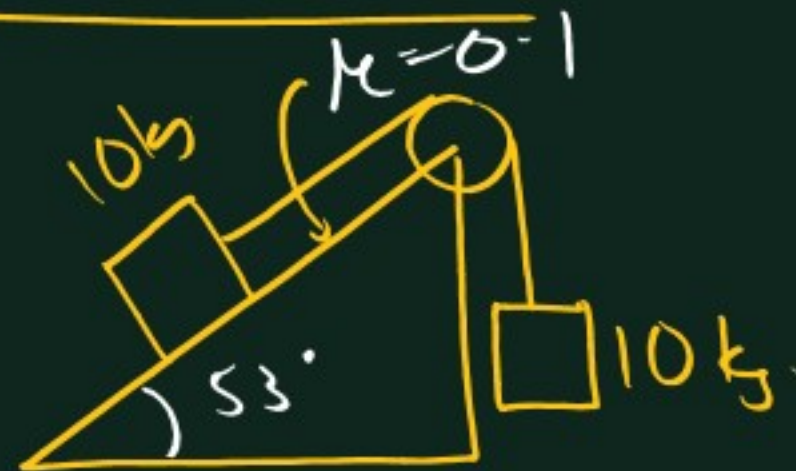
A) $f(10) < f_L(40)$

$\Rightarrow a = 0$

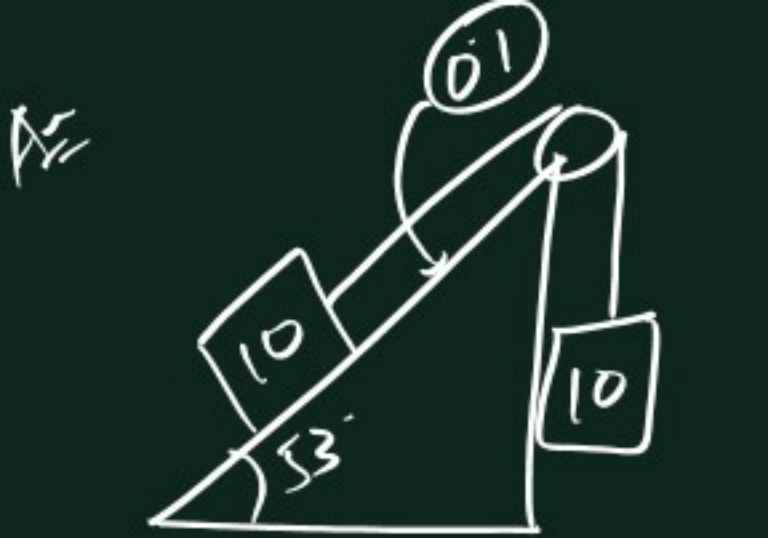
System at rest

$f = 10$ N up the incline.

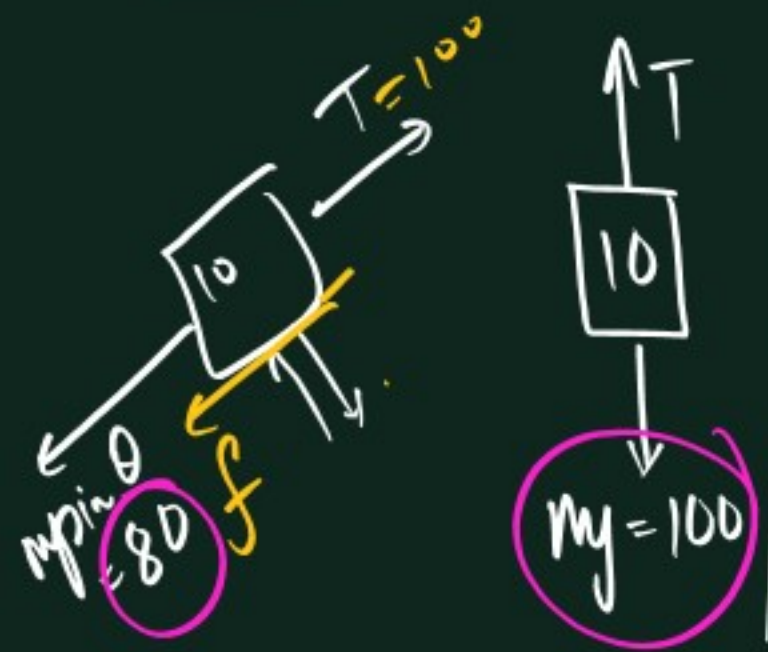
Q.



Find 'a' & 'f' ??



Assume rest



$$T = 100$$

$$\therefore f = 100 - 80 = \boxed{20}$$

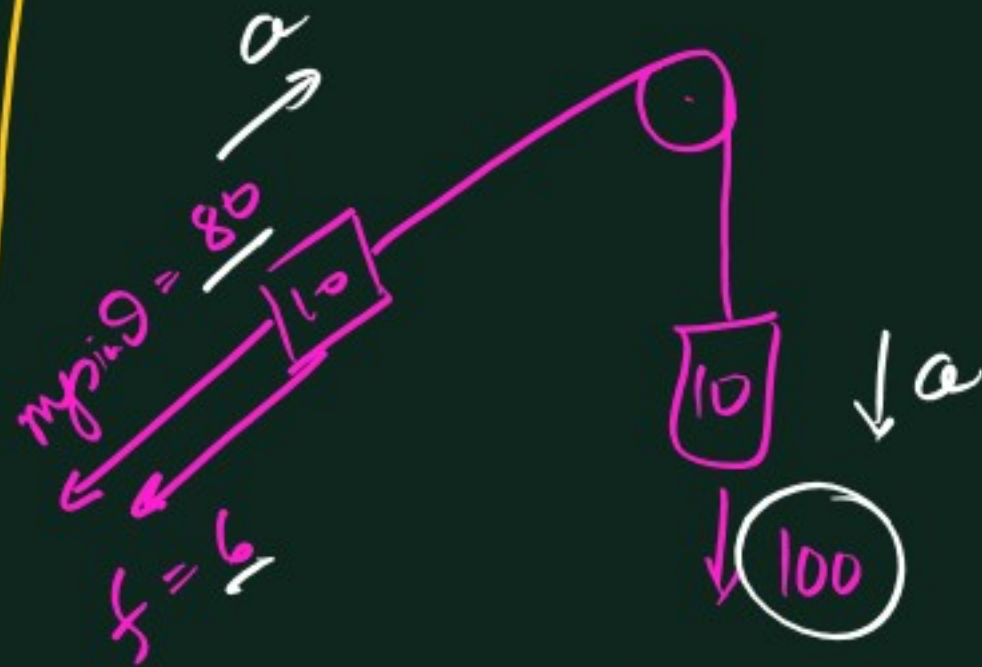
$$f_L = \mu (mg \cos \theta)$$

$$= 0.1 \times 60$$

$$= \boxed{6N}$$

$$\text{as } f > f_L$$

$\Rightarrow$ system moves a

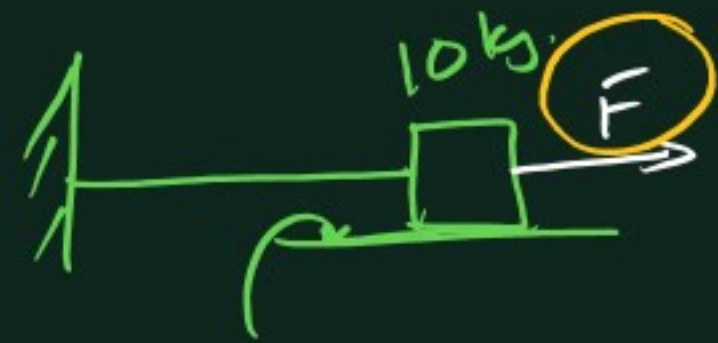


$$a = \frac{\text{Net pulling Force}}{\text{Total mass}}$$

$$= \frac{100 - 80 - 6}{20}$$

$$= \underline{0.7 \text{ m/s}^2}$$

$$\underline{f = 6N}$$



$$\mu = 0.4$$

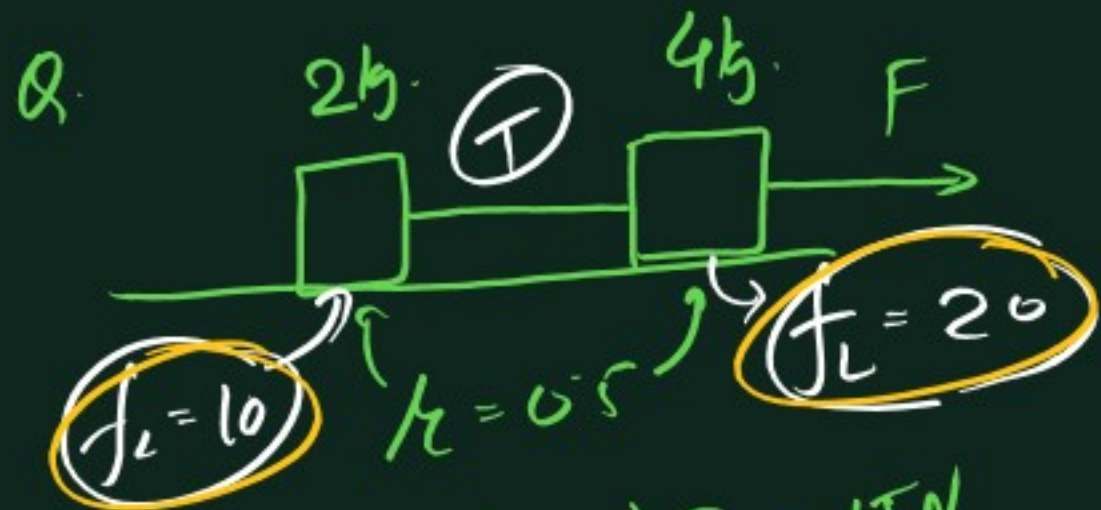
String at its normal length. Find T

$$\text{if } F = 20N.$$

$$f_L = \mu mg = \boxed{40N}$$

$$\text{As } F < f_L$$

$$\Rightarrow \underline{\underline{T = 0}}$$

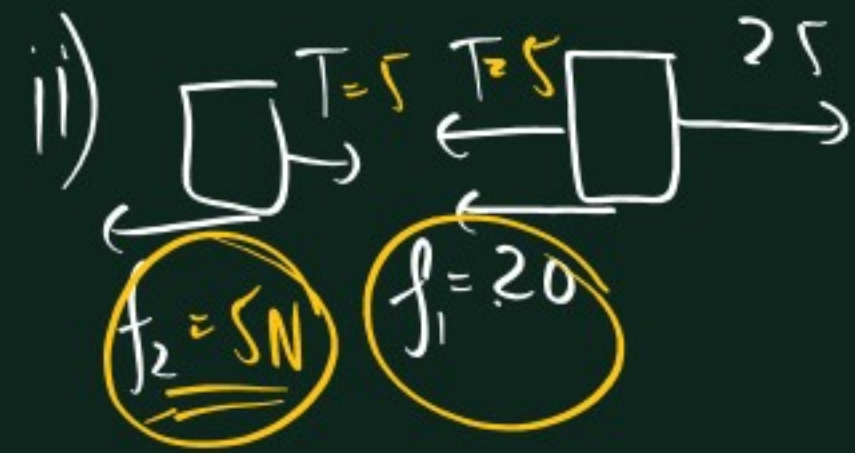


Find T if

- i) $F = 15\text{N}$
- ii) $F = 25\text{N}$
- iii) $F = 45\text{N}$

- A) 0 B) 5
C) 15 D) none

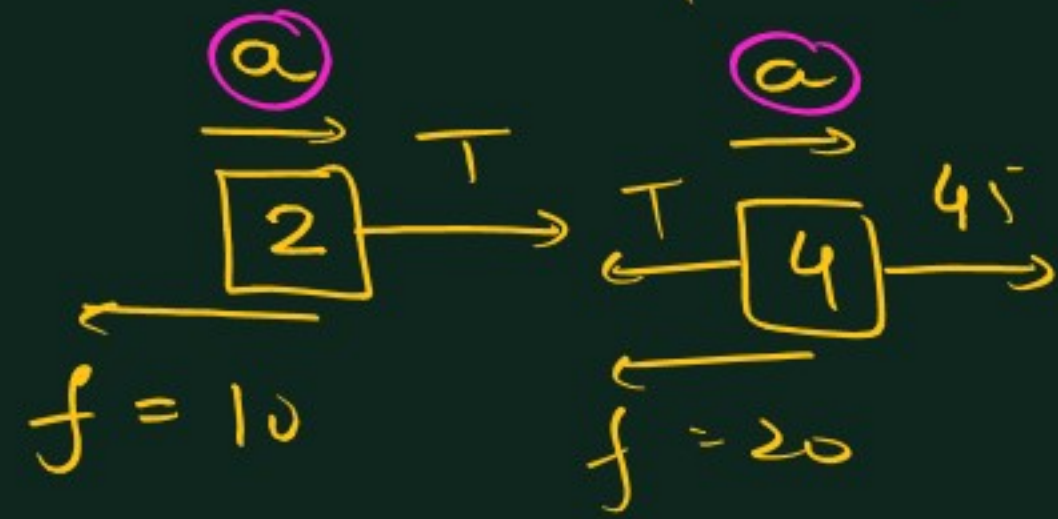
i) Until $F < 20\text{N}$, $T = 0 \Rightarrow T = 0$



$25\text{N} \Rightarrow 25 > (f_L)_1 \Rightarrow$ block tries to move

ii) If $F > (f_{L1} + f_{L2}) \Rightarrow$ 'a' (system moves)

Here $45 > (20 + 10)$

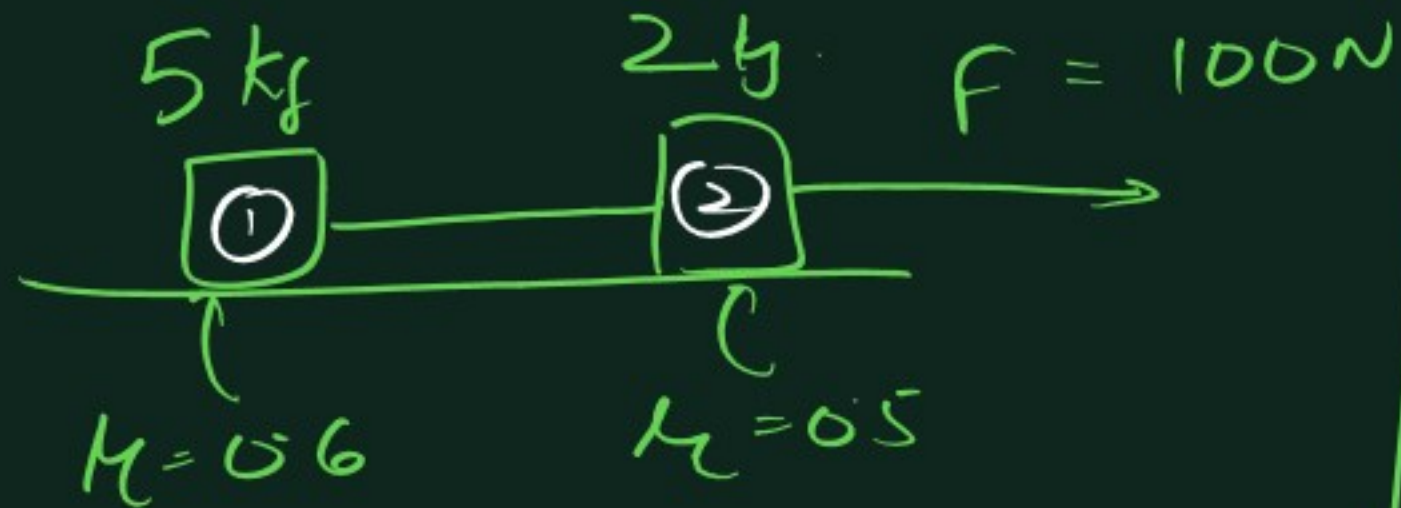


$$\left. \begin{aligned} T - 10 &= 2a \quad \text{--- (1)} \\ 45 - T - 20 &= 4a \quad \text{--- (2)} \end{aligned} \right\}$$

Add, $a = 2.5$
 $\therefore T = 15\text{N}$

String pulled
($T = 5\text{N}$)
 $\downarrow$
2kg pulled
by $T = 5\text{N}$
 $\downarrow$
 $f_2 = 5\text{N}$

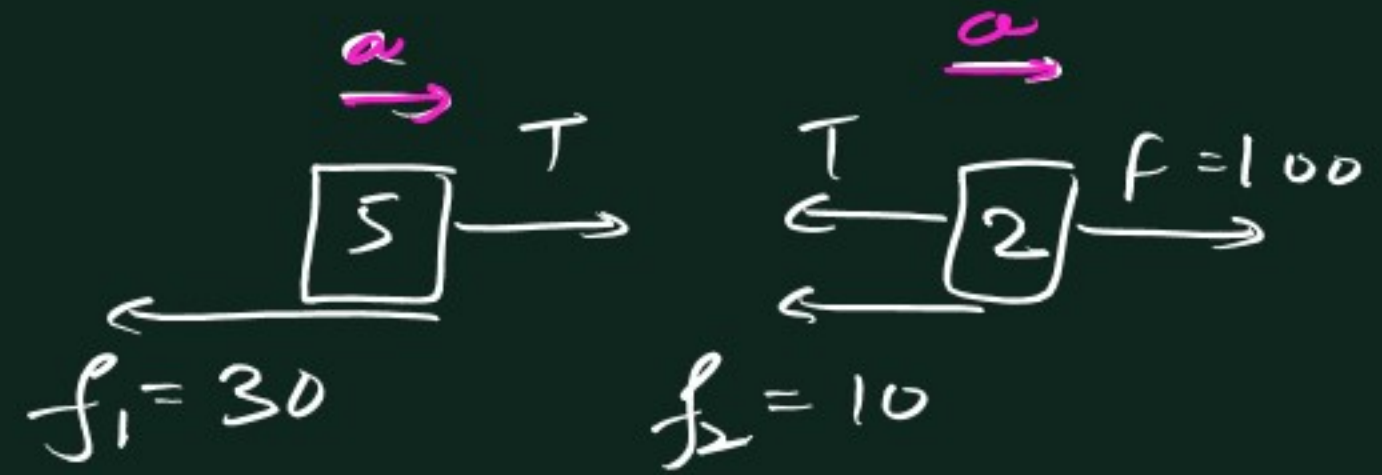
Q.



Find T, f_1, f_2

$$F > (f_{L1} + f_{L2})$$

$\begin{matrix} 100 \\ 6 \end{matrix} > \begin{pmatrix} \begin{matrix} 30 \\ 6 \end{matrix} + 10 \end{pmatrix}$



$$T - 30 = 5a \quad \text{--- (1)}$$

$$100 - T - 10 = 2a \quad \text{--- (2)}$$

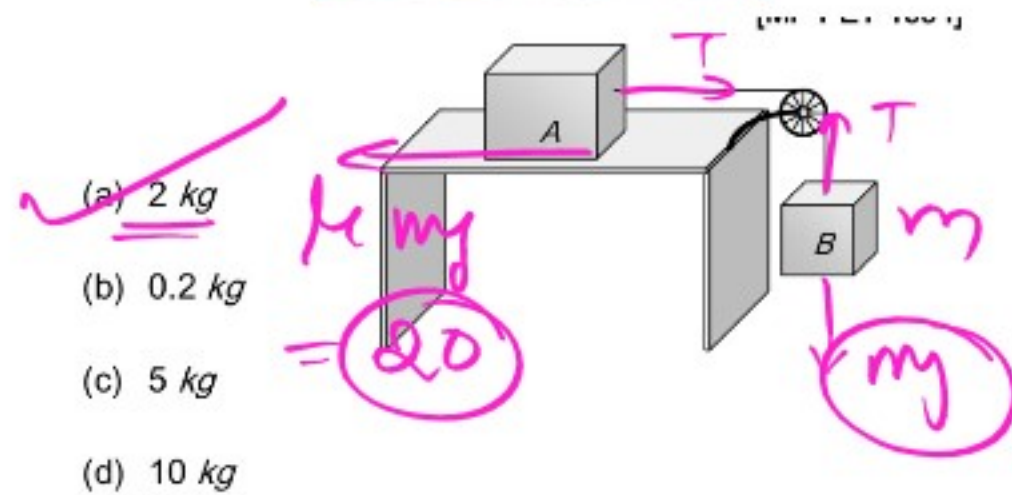
Add

$$100 - 30 - 10 = 7a$$

$$\Rightarrow a = \frac{60}{7}$$

$$\therefore T = 30 + 5 \times \frac{60}{7} = \frac{510}{7} \text{ N}$$

15. The blocks A and B are arranged as shown in the figure. The pulley is frictionless. The mass of A is 10 kg . The coefficient of friction of A with the horizontal surface is 0.20. The minimum mass of B to start the motion will be



- (a) 2 kg
(b) 0.2 kg
(c) 5 kg
(d) 10 kg

8. A 60 kg body is pushed with just enough force to start it moving across a floor and the same force continues to act afterwards. The coefficient of static friction and sliding friction are 0.5 and 0.4 respectively. The acceleration of the body is

- (a) 6 m/s^2
(b) 4.9 m/s^2
(c) 3.92 m/s^2
(d) 1 m/s^2

11. On a rough horizontal surface, a body of mass 2 kg is given a velocity of 10 m/s . If the coefficient of friction is 0.2 and $g = 10\text{ m/s}^2$, the body will stop after covering a distance of

[MP PMT 1999]

- (a) 10 m
(b) 25 m
(c) 50 m
(d) 250 m

13. A body of 10 kg is acted by a force of 129.4 N if $g = 9.8\text{ m/sec}^2$. The acceleration of the block is 10 m/s^2 . What is the coefficient of kinetic friction [EAMCET 1994]

- (a) 0.03
(b) 0.01
(c) 0.30
(d) 0.25

15. Consider a car moving along a straight horizontal road with a speed of 72 km/h . If the coefficient of kinetic friction between the tyres and the road is 0.5 , the shortest distance in which the car can be stopped is $[g = 10 \text{ ms}^{-2}]$

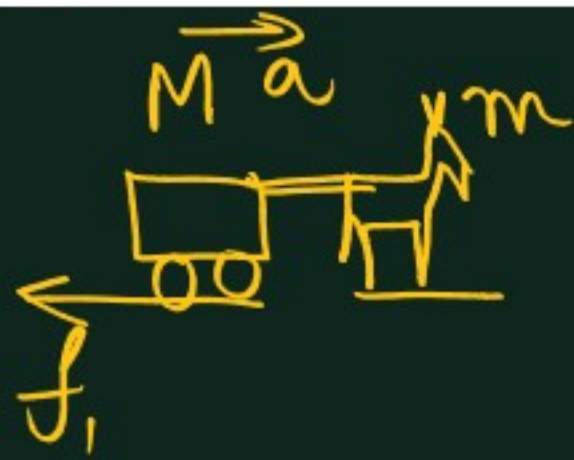
[CBSE PMT 1992]

- (a) 30 m (b) 40 m
(c) 72 m (d) 20 m

16. A 500 kg horse pulls a cart of mass 1500 kg along a level road with an acceleration of 1 ms^{-2} . If the coefficient of sliding friction is 0.2 , then the force exerted by the horse in forward direction is

[SCRA 1998]

- (a) 3000 N (b) 4000 N
(c) 5000 N (d) 6000 N



$$72 \times \frac{5}{18} = 20 \text{ m/s}$$

$$u = 20$$

$$f_k = \mu_k mg$$

$$\Rightarrow a = \frac{f_k}{m} = -\mu_k g = -5$$

$$v = 0$$

$$v^2 = u^2 + 2as$$

$$\Rightarrow 0 = 400 - 2 \times 5 \times s$$

$$\Rightarrow s = 40$$

$$\Sigma f = ma$$

$$\Rightarrow F_1 - f = Ma$$

$$\Rightarrow F_1 - \mu_k Mg = Ma$$

$$\Rightarrow F_1 = Ma + \mu_k Mg$$

$$= 1500(1 + 2) = 4500$$

29. If μ_s , μ_k and μ_r are coefficients of static friction, sliding friction and rolling friction, then [EAMCET (Engg.) 1995]

- (a) $\mu_s < \mu_k < \mu_r$ (b) $\mu_k < \mu_r < \mu_s$
(c) $\mu_r < \mu_k < \mu_s$ (d) $\mu_r = \mu_k = \mu_s$

32. A fireman of mass 60 kg slides down a pole. He is pressing the pole with a force of 600 N . The coefficient of friction between the hands and the pole is 0.5 , with what acceleration will the fireman slide down ($g = 10 \text{ m/s}^2$)

[Pb. PMT 2002]

- (a) 1 m/s^2 (b) 2.5 m/s^2
(c) 10 m/s^2 (d) 5 m/s^2

33. A block of mass $M = 5 \text{ kg}$ is resting on a rough horizontal surface for which the coefficient of friction is 0.2. When a force $F = 40 \text{ N}$ is applied, the acceleration of the block will be ($g = 10 \text{ m/s}^2$)

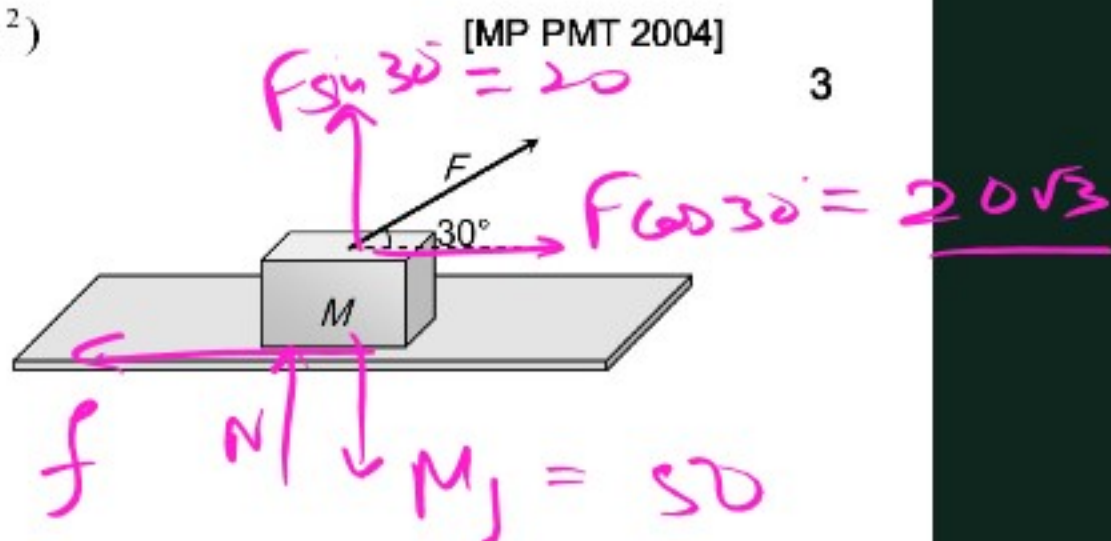
[MP PMT 2004]

(a) 5.73 m/sec^2

(b) 8.0 m/sec^2

(c) 3.17 m/sec^2

(d) 10.0 m/sec^2



10. When a bicycle is in motion, the force of friction exerted by the ground on the two wheels is such that it acts

[IIT 1990; Manipal MEE 1995; MP PET 1996]

- (a) In the backward direction on the front wheel and in the forward direction on the rear wheel
- (b) In the forward direction on the front wheel and in the backward direction on the rear wheel
- (c) In the backward direction on both front and the rear wheels
- (d) In the forward direction on both front and the rear wheels

wheels

$$N = M_j - 20 = 30$$

$$f_k = \mu N$$

$$= 0.2 \times 30$$

$$= 6 \text{ N}$$

$$\Sigma f = ma$$

$$\Rightarrow 20\sqrt{3} - 6 = 5a$$

$$\Rightarrow a = \frac{20\sqrt{3}}{5} - \frac{6}{5}$$

$$= 4\sqrt{3} - 1.2$$

$$= 6.9 - 1.2$$

$$= 5.7$$

$$40$$



$$\underline{f_1} - \underline{f_2} = (\underline{m + M}) a$$

$$f_1 = f_2 + (m + M) a$$

$$= 0.2 \times 1500 \times 10 + (2000) \times 1$$

$$= \underline{\underline{5000\text{N}}} \rightarrow \text{Force by Horse on earth / earth on Horse}$$

$$\begin{aligned} & \mu(M + m) g \\ &= 0.2 \times 2000 \times 10 \\ &= \underline{4000\text{N}} \end{aligned}$$